

# **RMUTT HPE Aruba Networking campus associate**

**Aruba Campus Network Training • 26 ส.ค. 2569**

# **Agenda (1/3)**

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**Overview Aruba Solution**

**Network devices & roles**

**Campus architecture**

**CX / Gateway / AP portfolio**

## **Agenda (2/3)**

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### **Module 1 : Aruba Switch**

**Switch series, stacking, PoE**

**Lab 1: Device management**

**Lab 2: LAG & VLAN**

**Lab 3: Layer 3 Routing**

**Knowledge check?**

## **Agenda (3/3)**

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### **Module 2 : Controller + AP**

**Controller & AP onboarding**

**WLAN: PSK & MAC auth**

**WLAN: 802.1X**

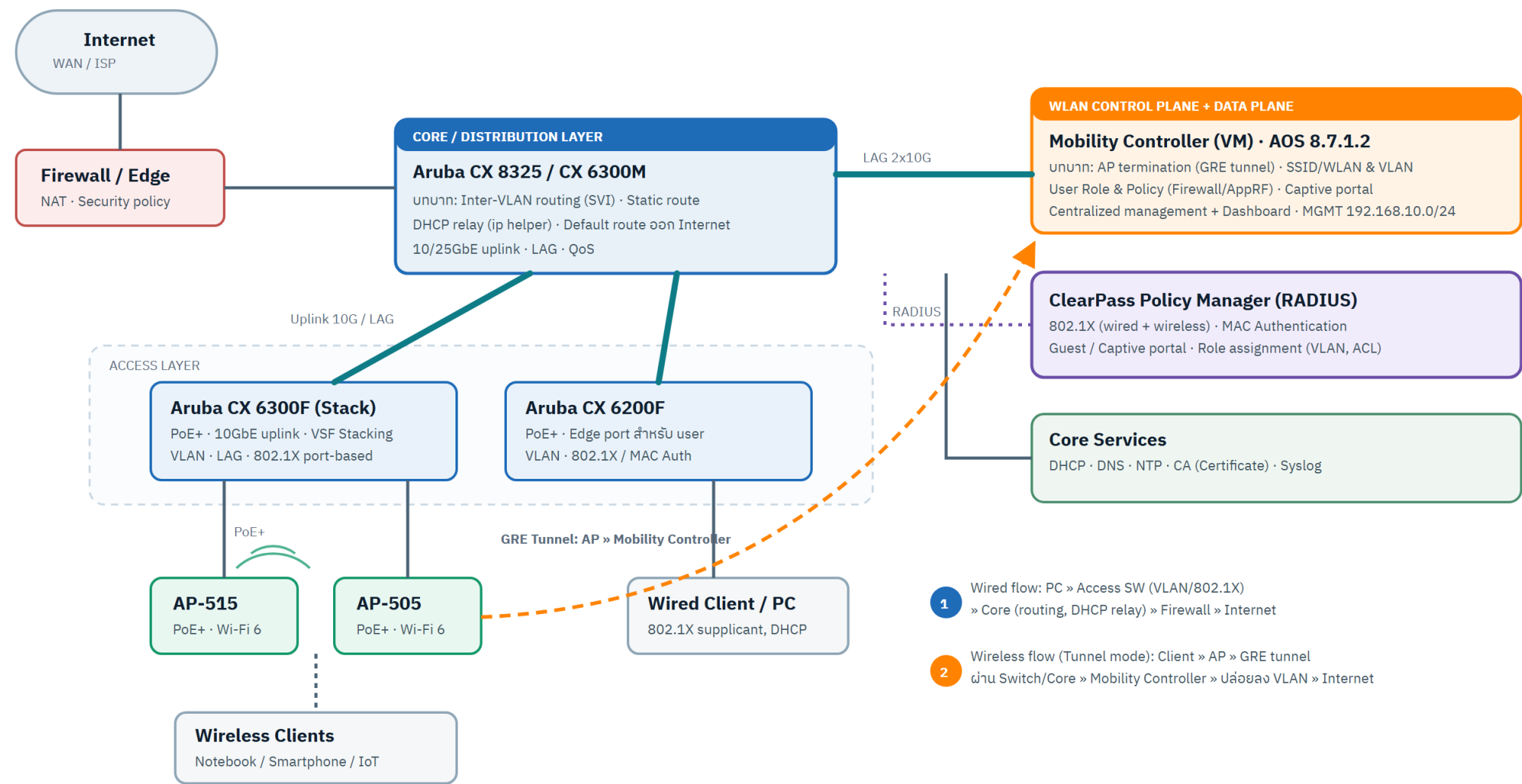
**WLAN: Captive portal (Guest)**

**Role & Policy · Q & A**



# Aruba Campus Architecture – Overview

Switch + Controller + AP · Device roles & traffic flow — RMUTT / Commserv Siam · 26 ส.ค. 2569



# Network Devices

# Networking devices: switch

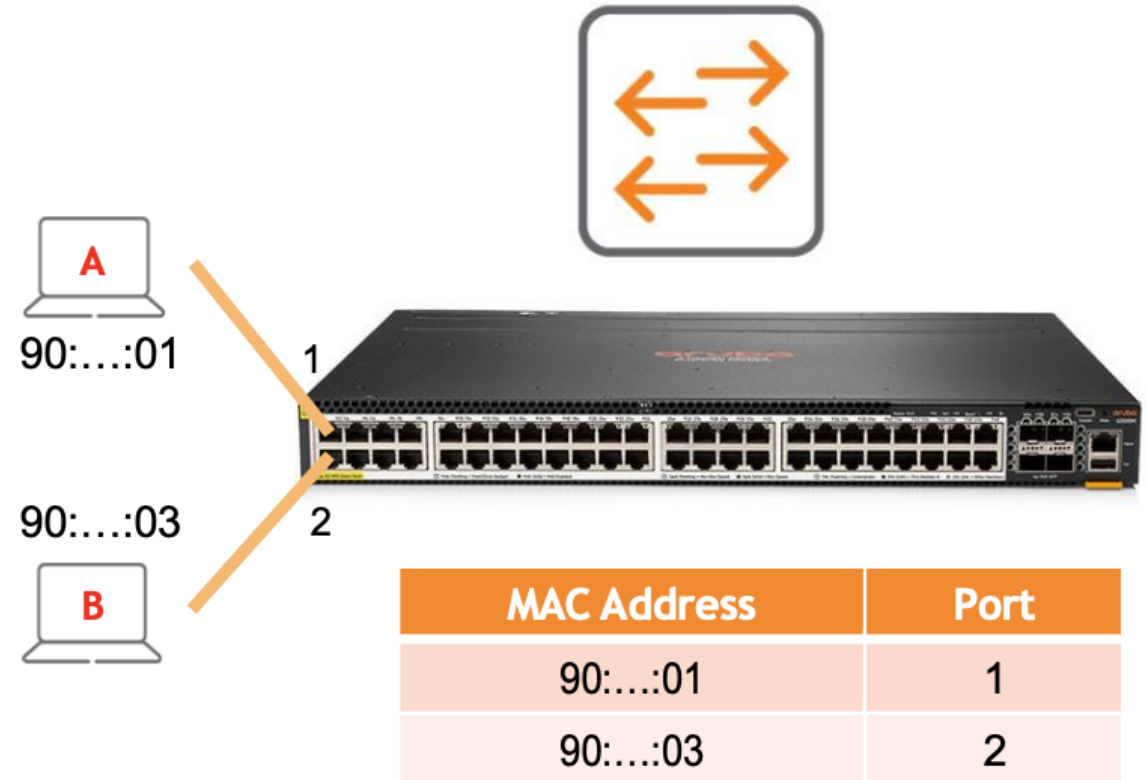
L2 device: Forwards Ethernet frames based on destination MAC addresses

## Multiple ports connect endpoints

- PCs, printers, cameras, and more
- Commonly 8, 24, 48, or more physical ports
- Transparent to endpoints

## Special L2 protocols

- Performance, reliability, security
- STP, LLDP, 802.1Q



# Networking devices: router

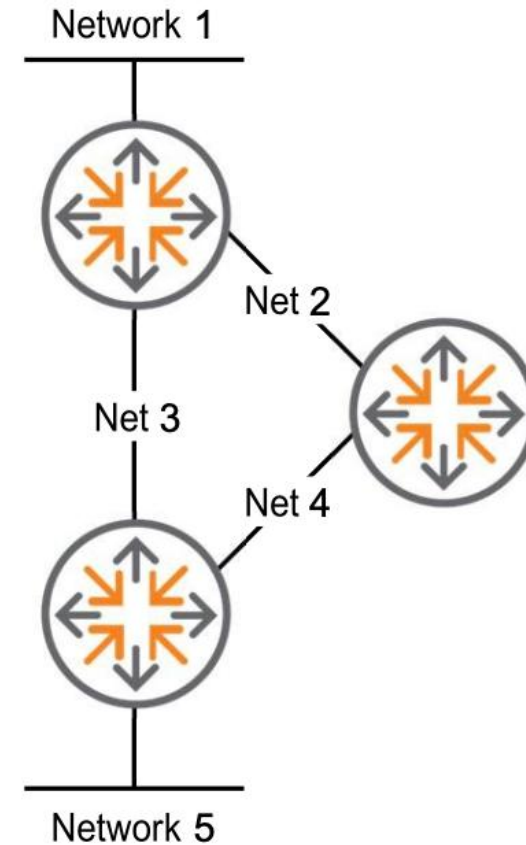
Layer 3 device: Forwards packets based on destination IP address

## Functionality and features

- Connect separate networks into an inter-network
- Offer WAN connectivity to networks

## L3 routing protocols

- Learn all possible paths, choose a best path
- RIP, OSPF, BGP





# Networking devices: multi-layer switch

Performs both L2 switching and L3 routing

## L3 Routing

- Internal routing functionality
- Runs RIP, OSPF, BGP

## L2 Switching

- Multiple ports connect endpoints
- Uses STP, LLDP, VLANs



# Networking devices: access point

Enables wireless users to access wired resources and roam about

## Functionality

- Bridges wireless devices and wired networks.
- Transform Ethernet into Wi-Fi frames and back



## Varieties of APs

- Internal or external antennas
- Single or Dual Ethernet Ports
- Indoor or outdoor



# Networking devices: firewall

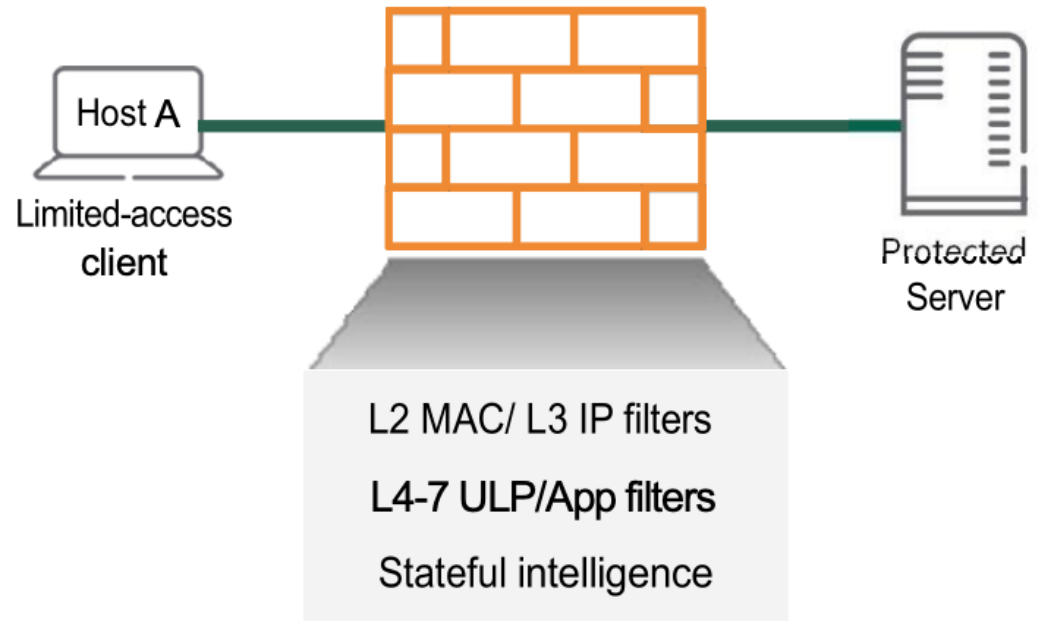
L2-7 filtering and security

## Functionality

- Block unauthorized/inappropriate access
- Permit authorized/appropriate access

## Types

- Stateless vs Stateful
- Functionality built into switches, gateways





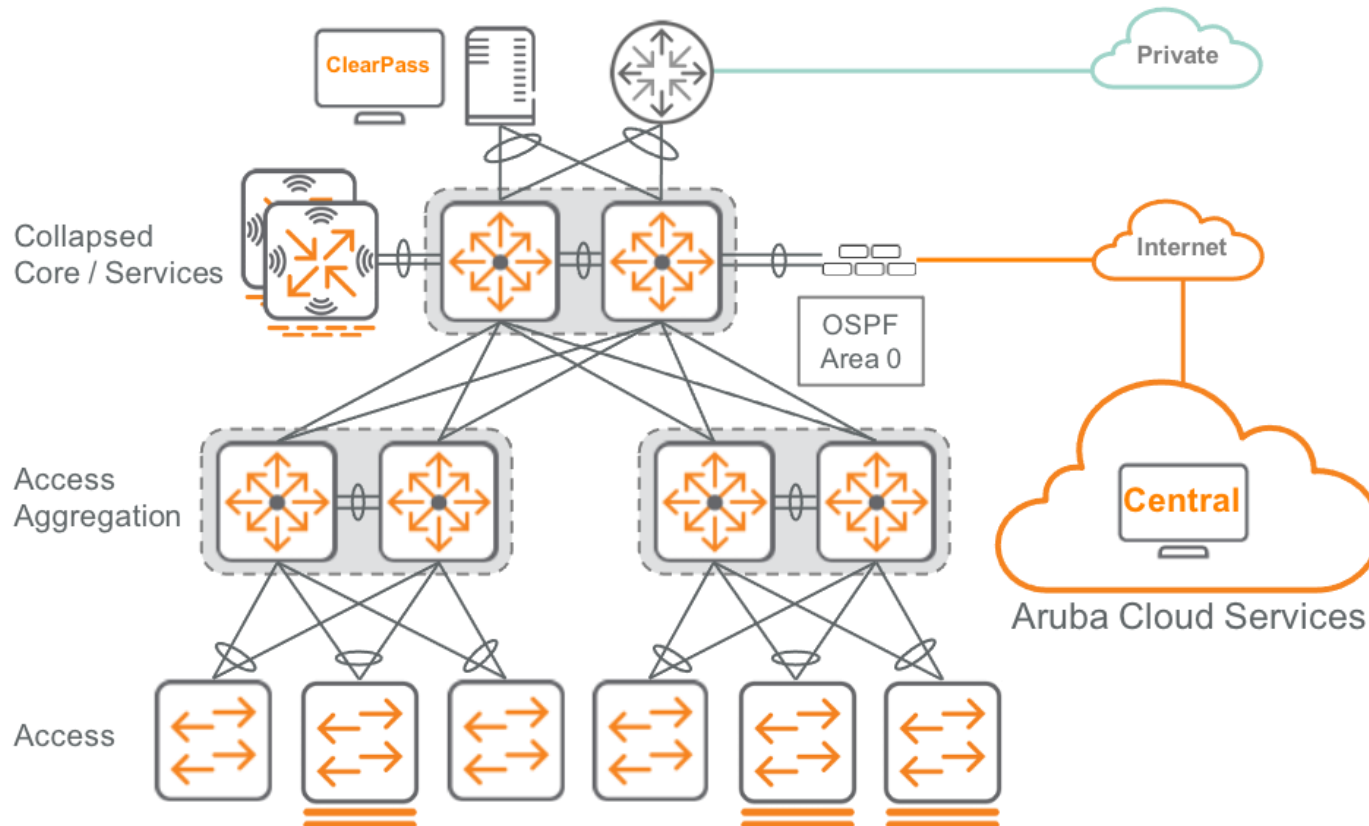
# Introduction switch

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# CX switching solutions for all networks

For branch, campus, mid and large enterprises, SMB, and data center networks



**Data Center**

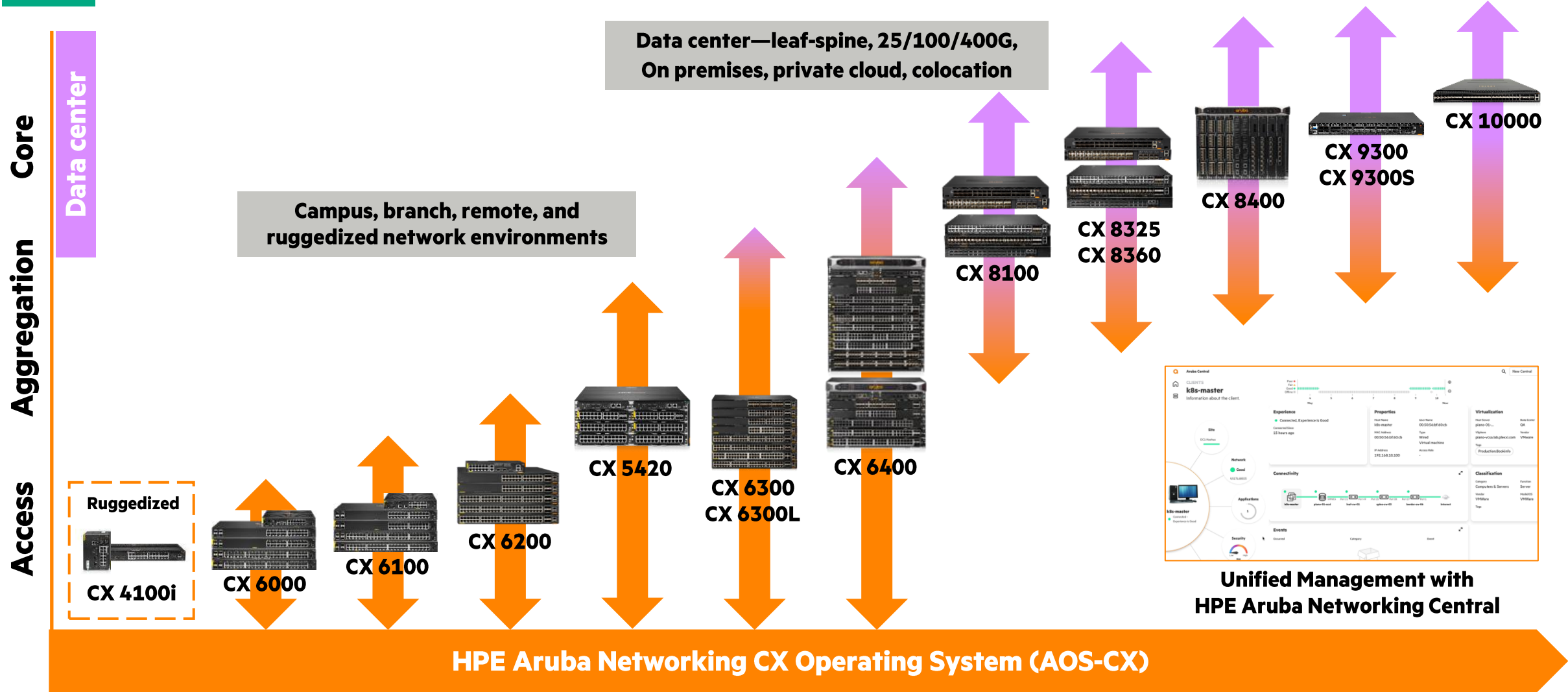
**Enterprise Core**

**Aggregation and Distribution**

**Campus and Branch Access**

**Ruggedized**

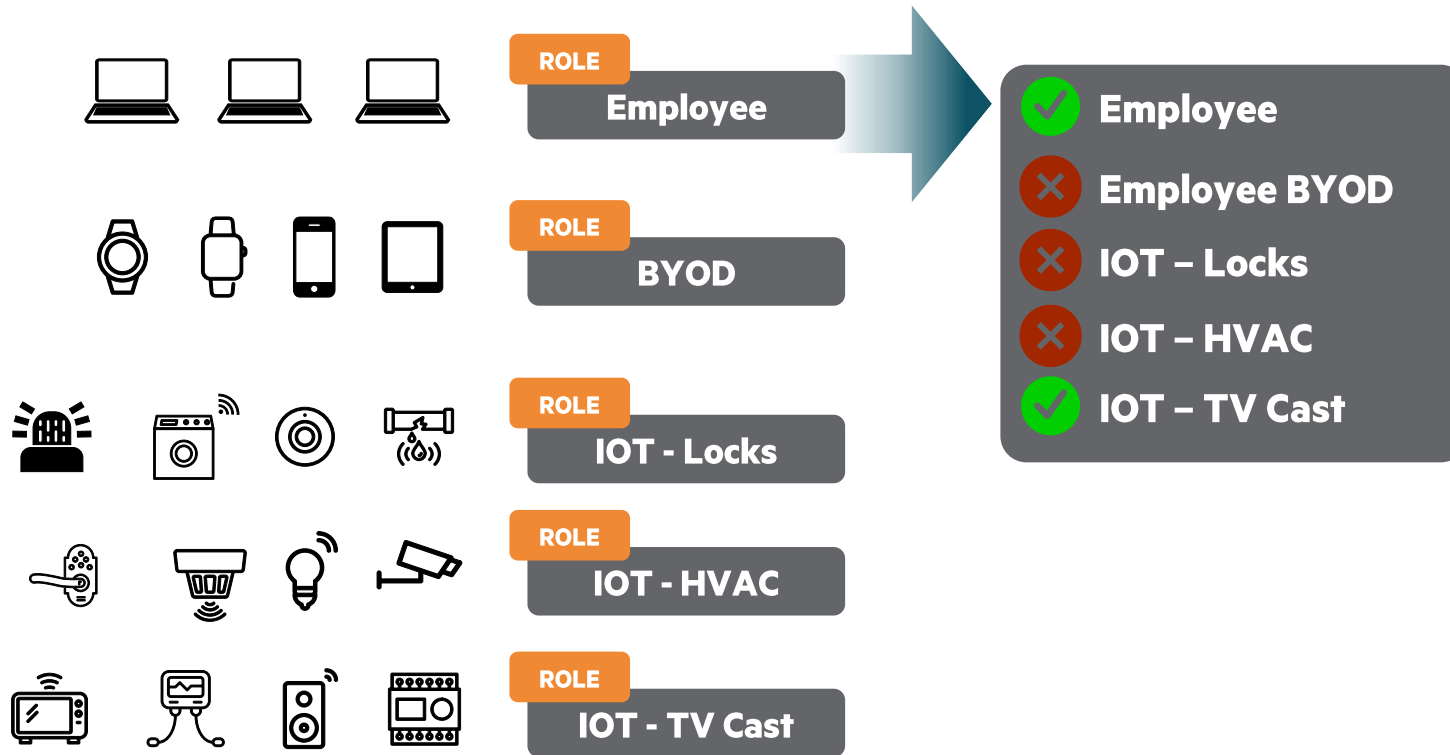
# HPE Aruba Networking CX switch portfolio



One common operating model from campus edge to DC



# HPE Aruba Networking User Roles



## DYNAMIC SEGMENTATION

Software defined approach eliminates VLAN sprawl and simplifies policy implementation

Delivers wired, wireless, and SD-WAN micro-segmentation needed for securing end-user and IoT devices

# AP platforms for comprehensive campus and branch connectivity

Wi-Fi 7

Wi-Fi 6E

Wi-Fi 6

AOS-10  
BASED



## Mid-range Wi-Fi 7 730 Series (AP-734/735)

Tri-band support  
2x2:2SS (8RU), 9.4Gbps,  
2x 5GE, 2xUSB, embedded GPS &  
barometer, 2 x IoT radios  
Ultra tri-band filtering

AOS-10  
BASED



## Flagship Wi-Fi 7 750 Series (AP-734/735)

Tri-band support  
4x4:4SS (8RU), 18.8Gbps,  
2x 10GE, 2xUSB, embedded GPS &  
barometer, 2 x IoT radios  
Ultra tri-band filtering



## Entry level Wi-Fi 6E 610 Series (AP-615)

Dual-radio/tri-band  
2x2:2SS (8RU), 3.6Gbps,  
1x 2.5GE, USB, embedded GPS



## Mid-range Wi-Fi 6E 630 Series (AP-634/635)

Tri-band support  
2x2:2SS (8RU), 3.9Gbps,  
2x 2.5GE, USB, embedded GPS, UTB



## Flagship Wi-Fi 6E 650 Series (AP-654/655)

Tri-band support  
4x4:4SS (8RU), 7.8Gbps,  
2x 5.0 GE, USB, embedded GPS, UTB



## Entry level Wi-Fi 6 500 Series (AP-503)

Dual-radio 2x2:2SS, 1.5Gbps, 1x  
1GE, USB



## Entry level Wi-Fi 6 500 Series (AP-505/504)

Dual-radio 2x2:2SS (8RU), 1.5Gbps, 1x  
1GE, USB, DC, BLE/Zigbee, IPM, FTM



## Mid-range Wi-Fi 6 510 Series (AP-515/514)

Dual-radio 2x2:2SS (2.4GHz), 4x4:4SS (5GHz),  
2.7Gbps, 1x2.5GE + 1xGE, USB, DC, BLE/Zigbee,  
IPM, FTM



## High-performance Wi-Fi 6 530 Series (AP-535/534)

Dual-radio 4x4:4SS, 3.98Gbps, 2x 5GE, USB,  
DC, BLE/Zigbee, IPM, FTM



## Flagship Wi-Fi 6 550 Series (AP-555/554)

Dual-radio 4x4:4SS (2.4GHz) 4x4:4SS x2 or  
8x8:8SS (5GHz), 5.4Gbps, 2x 5GE, USB, DC,  
BLE/Zigbee, IPM, FTM

< Low-density to high-density, demanding environments >

# Always-on, always available enterprise networking

Resilient, scalable architecture that eliminates downtime for business continuity

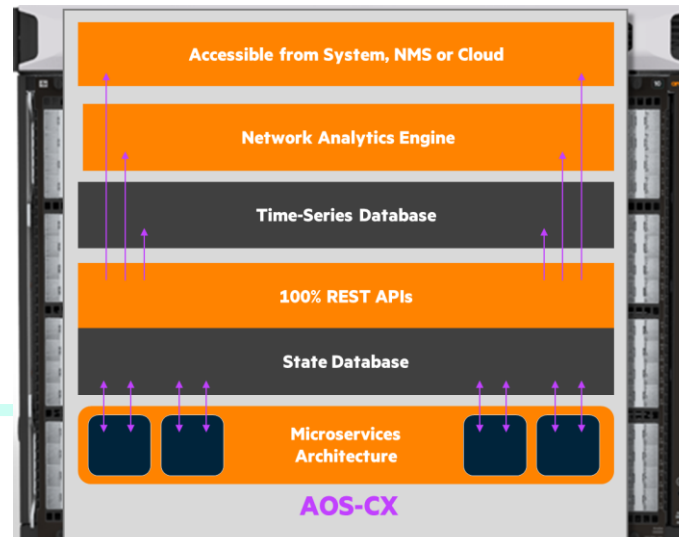


Resilient networks are designed to remain always-on, ensuring uptime for **business continuity** when every moment matters.

- Management and operations
- Physical platform redundancy
- Physical design redundancy
- Logical design redundancy
- Disaster recovery
- Software upgrades

## Resilient by design

AOS-CX is a modern, cloud-native, and modular operating system built to be database-centric with independent micro-services and state synchronization.



VSX Live Upgrades

Hot Patching

VSF ISSU

Built-in analytics

Rollout validated configs

Clusters & Stacking

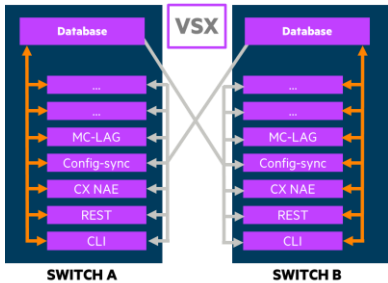
Hot-swappable parts

Always-on PoE

# CX creates business continuity across the network thru SW and HW

Resilient, scalable architecture by design

## VSX Live Upgrades



ISSU orchestrated upgrades to eliminate maintenance windows

## Rollout validated configs

Automated rollouts with NetEdit or Central Multi-Editor

## VSF ISSU & hot patching

Minimize downtime during upgrades across a stack

## Clustering & Stacking

Grow quickly, simplify mgmt., extend distances with long range trvrs.

## Built-in analytics

Visibility and alerts for faster troubleshooting and MTTR

## Always-on PoE

Continuous power for IoT and APs during switch updates

## Redundant and hot-swappable parts

Power supplies, fans, management, and fabric



# Virtual Switching Framework (VSF)

**For resilient and scalable campus and branch access networks**

**Simplified Management** with virtual switch stack

**Grow** your network quickly

**Go further with stacking** across campus and site using long range transceivers

**Fast stack with CX Mobile App** that auto-detects potential stack members for a virtualized stack with just a few taps

**Configure a stack quickly** within Central UI



## CX 6300

10-unit CX 6300 stack

Flexibility to combine 6300M and 6300F models

Uses 10/25/50 GbE uplinks

## CX 6200

8-unit CX 6200 stack

Flexibility to combine 6200M and 6200F models

Uses 10GbE uplinks

\* 6300L only stacks with other 6300L models.  
6200 and 6300 mixed stack not allowed

# In-Service Software Upgrades (ISSU) with VSF Stacking

Rapid, interruption-less software upgrades for stacked CX 6300 switches<sup>1</sup>

## What is an ISSU?

- Software upgrades within a release, without traffic interruption. The data plane will keep forwarding traffic during entire upgrade process
- Requires conductor and standby as part of the VSF stack.
- Supports up to 10-member stack

## Benefits

- Increase availability during upgrade
- Data plane is not affected and keeps forwarding traffic
- Maintain end-points and application connectivity through the upgrade process
- Includes bug/vulnerability fixes and new features

## Total Downtime: 0%

Customer starts ISSU, and observes system ready validation

Standby and members update, then connect to standby database

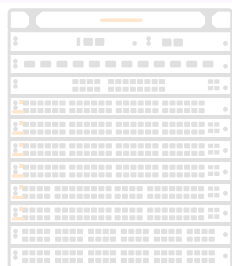
Database sync from active conductor to standby occurs

Switchover from active conductor to standby, new active takes over

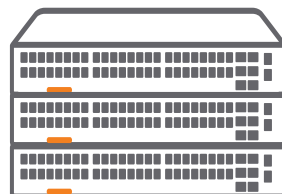
Former active updates & becomes new standby without reboot

<sup>1</sup> VSF ISSU not available on CX 6300L switch models

Configurations  
Supported



Chassis



Stack



Standalone

# CX 6200 Switch Series

Lite Layer 3 stackable access switches for branch offices and SMBs

Flexible growth with stackable gigabit connectivity with fast 1/10G uplinks

Power IoT & Wi-Fi devices with up to 1440W Class 4 to Class 6 PoE

Scalable growth with **8 member VSF stacking**

Real-time monitoring and troubleshooting with **Network Analytics Engine (NAE)**

Simple and secure access with **Dynamic Segmentation** that automates user and device policy

## CX portfolio benefits

Advanced configs and programmability with AOS-CX

Single pane of glass Central across wired, wireless, WAN

Flexible management via Web GUI, CLI, NetEdit, Central

Limited lifetime warranty and options for adv.TAC support

**11**

Fixed switches  
(power & fans)

**5**

Modular switches  
(power & fans)

**4**

Built-in 1 and 1/10 GbE uplinks



# CX 6300 Switch Series

Powerful Layer 3 stackable switches ready for demanding environments

Flexible growth with stackable gigabit connectivity with fast **10/25/50GbE<sup>1</sup> uplinks**, high density SFP+

Built for IoT and Wi-Fi 6 with up to **2880W Class 6 and Class 8 PoE** and HPE Smart Rate **multi-gigabit** Ethernet

Real-time monitoring and troubleshooting with **Network Analytics Engine (NAE)**

Simple and secure access with **Dynamic Segmentation**; campus and branch fabric with **VXLAN and BGP EVPN**

Power-to-port airflow for cost-effective **1GbE ToR and OOBM data center** deployments

## CX portfolio benefits

Advanced configs and programmability with AOS-CX

Single pane of glass Central across wired, wireless, WAN

Flexible management via Web GUI, CLI, NetEdit, Central

Limited lifetime warranty and options for adv.TAC support

**4**  
Fixed power switches

**12**  
Modular power switches

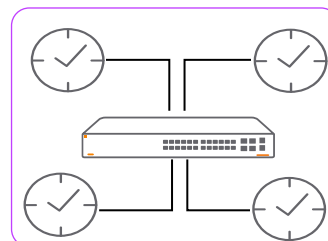
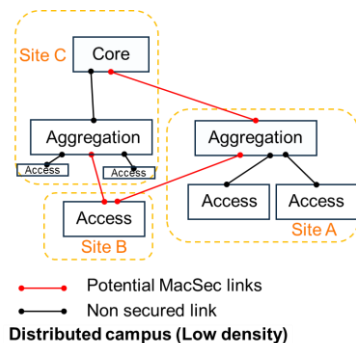
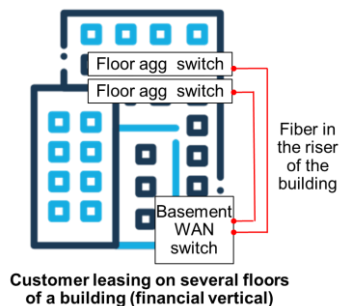
**2880W**  
60W and 90W PoE



**10 member**  
VSF stacking



# Bringing performance and versatility to campus networks

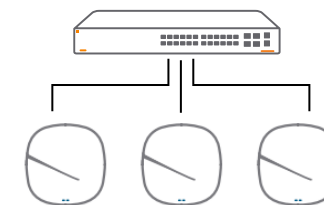


Synchronization and validation across systems

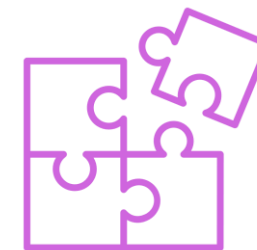


Video conferencing and multimedia consumption

**Time synchronization** and Audio Video Bridging support with precision time protocol (PTP) boundary clock (BC) and transparent clock (TC).

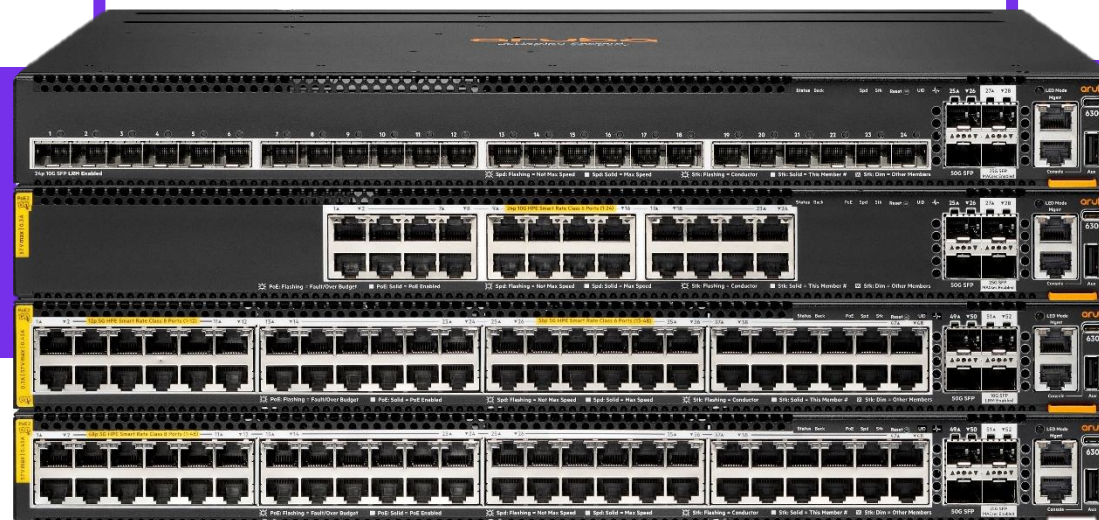


Next-gen 802.3bt enabled wireless access points



Internet of things devices and systems, like smart lighting

**Stronger data link layer** security with standards-based MACsec 256 encryption and authentication on uplinks and downlinks.



**More performance, power, and connectivity options** with up to 10G Smart Rate, Class 8 (90W) PoE, and LRM support.



# HPE Aruba Networking CX Switches – End-to-end portfolio

## Always-on Reliability

- HA with ISSU, HW, SW resiliency
- Proactive analytics

## Switching Solution for Any Network

Strong, scalable switching portfolio for branch, campus, mid and large enterprises, SMB, and data center networks

## Secure Access

- Dynamic Segmentation
- Campus- branch fabric EVPN/VXLAN
- NetConductor

## Simplified Operations

- Single OS from access to core to DC
- Smart automation for flawless configs with NetEdit
- Built-in analytics with deep visibility and clear alerts
- Management flexibility (Cloud, on-prem, Web GUI, CLI, NetEdit, AFC)
- Unified cloud-based single pane of glass management for wired, WLAN, SD-WAN

## HPE Aruba Networking CX Switches



## Technology Innovation

- AOS-CX: Programmable, modern, designed to scale access to DC
- NAE: built-in analytics monitor, baseline, troubleshoot
- NetEdit: error-free automated rollouts
- VSX live upgrades, VSF rolling stack, & Hot Patching: HA during upgrades
- In-house ASICs: high-performance, programmable agility
- Central: Unified, AIOps, NetConductor

## Faster Troubleshooting

- NAE built-in for monitoring, alerts, troubleshooting and root cause analysis
- Central integration with AI insights

## Scalable, High-Performance Enterprise Switching

High-density fixed and modular switches with non-blocking architecture, speeds from 1G to 400G, extended table sizes, high power PoE, multi-gig, 80 Plus power supplies, fiber and copper transceivers and DACs

## Customer First, Customer Last

- Industry leading warranty
- Global support and services
- Consumption choice: GL4A, NaaS
- Financing: HPEFS



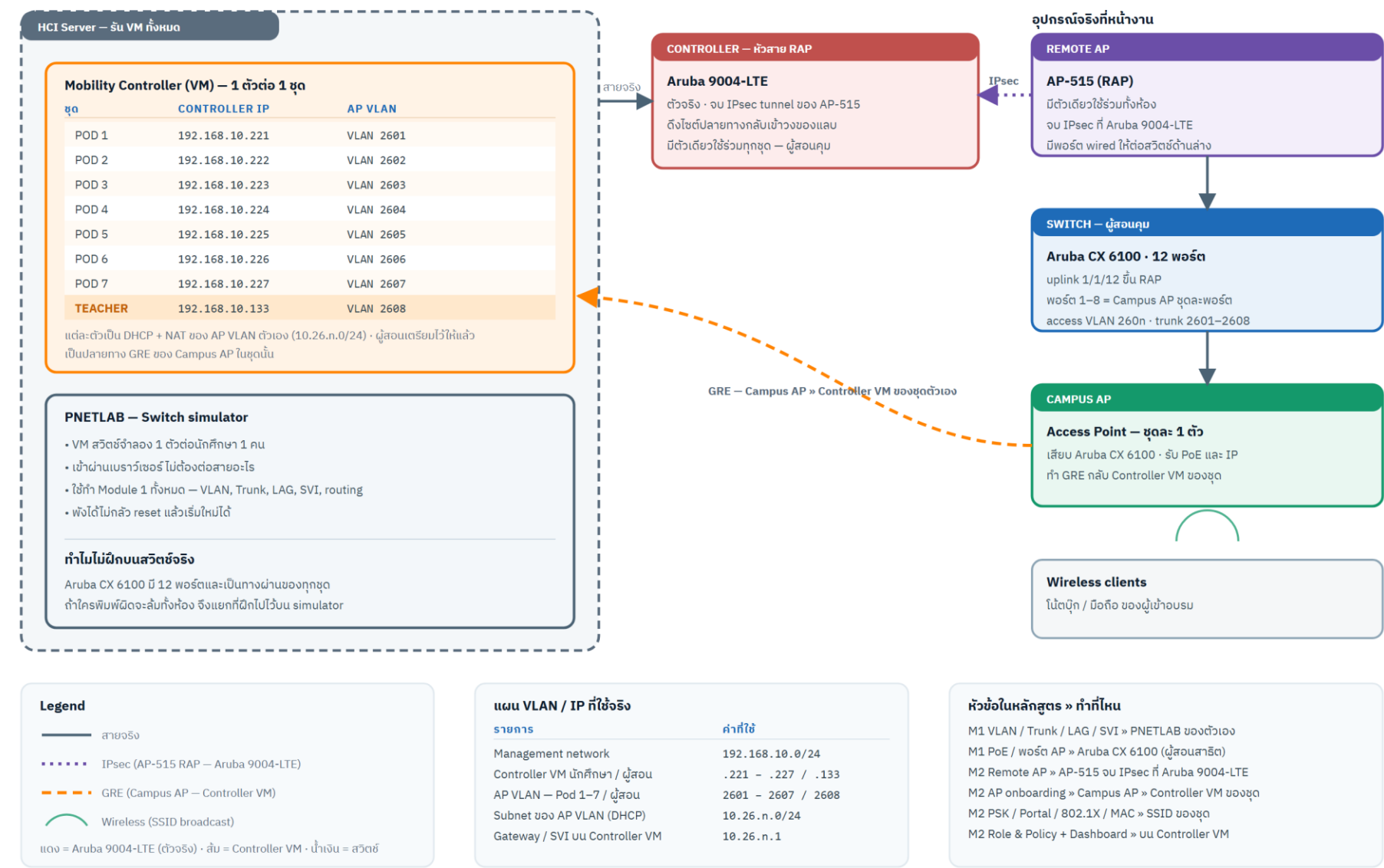
# Switch Labs

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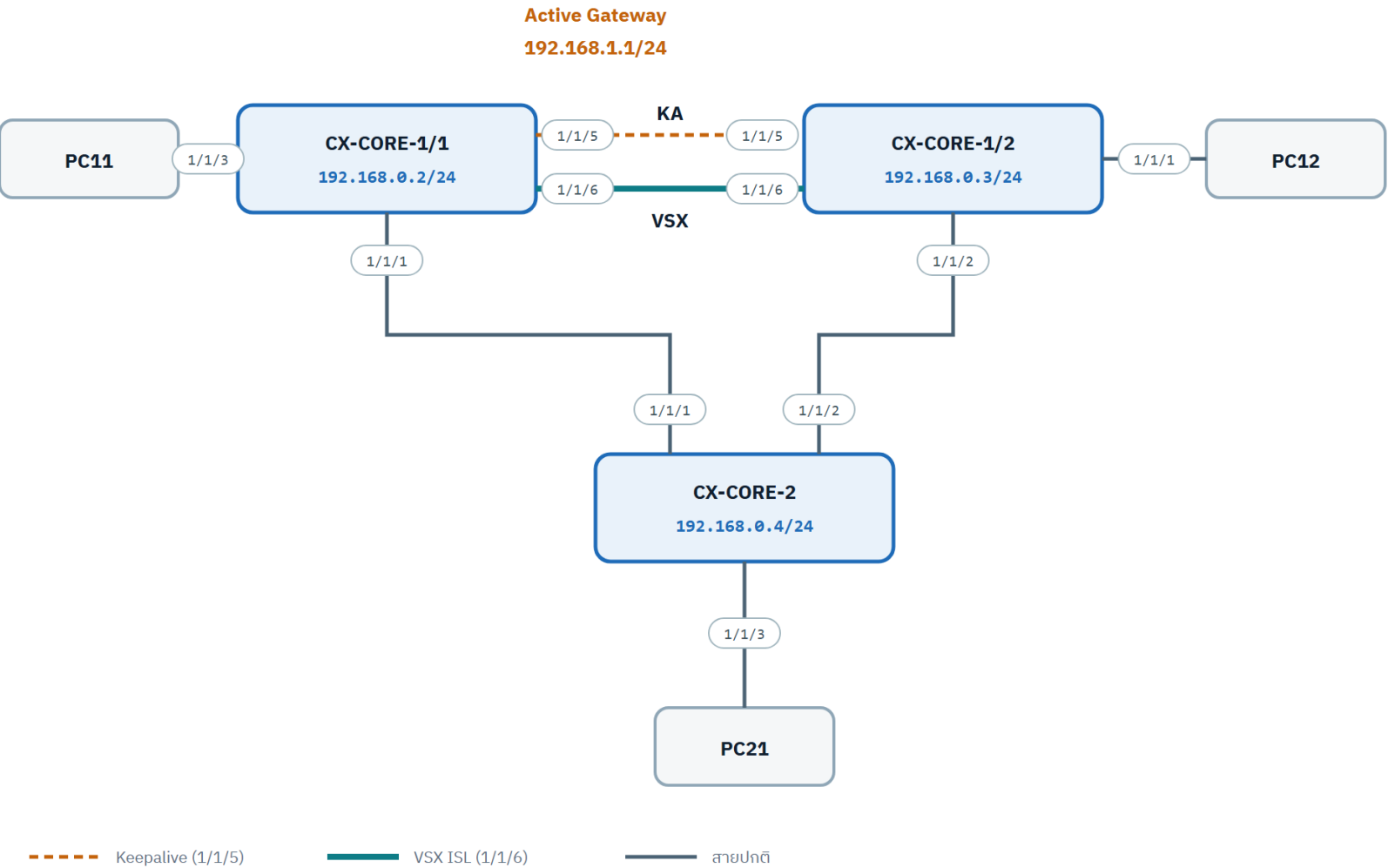
# Aruba Campus Network Training — Lab Topology

RMUTT x Commserv Siam · 26 ส.ค. 2569 · RAP จบที่ Aruba 9004-LTE · Campus AP จบที่ Controller VM ของชุดตัวเอง · 7 ชุดนักศึกษา + 1 ชุดผู้สอน



# Switch Lab uu PNETLAB

RMUTT x Commserv Siam · 26 ส.ค. 2569 · VSX pair + Active Gateway · สวิตช์จำลอง 3 ตัวและ PC 3 เครื่องต่อชุด



## การใช้งาน PNETLAB

URL <http://10.69.11.10/>  
ผู้ใช้ pod1 - pod7 (ตามชุดตัวเอง)  
รหัสผ่าน P@ssw0rd

ทุกคนในชุดใช้ชื่อกันเดียวกัน — ตกลงกันก่อนว่าใครกดอะไร

## ค่าที่ใช้ในผังนี้

| รายการ                    | ค่า                       |
|---------------------------|---------------------------|
| CX-CORE-1/1 · CX-CORE-1/2 | 192.168.0.2 · 192.168.0.3 |
| CX-CORE-2                 | 192.168.0.4               |
| Active Gateway (VSX pair) | 192.168.1.1/24            |
| VSX ISL                   | 1/1/6 ↔ 1/1/6             |
| VSX Keepalive             | 1/1/5 ↔ 1/1/5             |
| ลงโป CX-CORE-2            | 1/1/1 และ 1/1/2           |

## อ่านผังนี้อย่างไร

- สวิตช์สามตัวเป็น VM คนละชุดกับเพื่อน — พังได้ reset ใหม่ได้
- CX-CORE-1/1 กับ CX-CORE-1/2 จับคู่เป็น VSX
- CX-CORE-2 มองขึ้นไปเห็น VSX pair เป็นสวิตช์ตัวเดียว
- PC11 » 1/1/3 · PC12 » 1/1/1 · PC21 » 1/1/3

## LAB 0 เข้า PNETLAB และเปิดเครื่องมือให้ครบ

ที่โต๊ะ: เข้าห้องแล็บของชุดตัวเองได้ เห็นสวิตช์ครบสามตัว และเข้า Controller VM ของชุดได้

1. เปิด <http://10.69.11.10/> แล้วล็อกอินด้วยผู้ใช้ของชุดตัวเอง — POD 1 ใช้ pod1 รหัส P@ssw0rd
2. เปิด lab ของชุดตัวเอง กด START แล้วรอให้ทั้งสามสวิตช์ขึ้น
3. คลิกที่สวิตช์เพื่อเปิด console — เริ่มที่ CX-CORE-1/1
4. เปิดอีกแท็บ เข้าเว็บ Controller VM ของชุดตัวเอง เช่น POD 1 » <https://192.168.10.221:4343>
5. จดไว้: สวิตช์ 192.168.0.2 / .3 / .4 · Active Gateway 192.168.1.1 · Controller 192.168.10.22n

### ตรวจสอบ

```
show version
show interface brief
```

### ระวัง

ทุกคนในชุดใช้ล็อกอิน PNETLAB เดียวกัน — ถ้าสองคนกดพร้อมกันจะแย่ง console ตกลงกันก่อนว่าใครทำตัวไหน



## LAB 1 ตั้งชื่อและ IP ให้สวิตช์ทั้งสามตัว (1/2)

PNETLAB (สวิตช์จำลอง) แยกออกว่ากำลังพิมพ์อยู่ที่สวิตช์ตัวไหน และ ping ถึงกันได้ทั้งสามตัว

1. ที่ CX-CORE-1/1 ตั้งชื่อและ IP บนพอร์ตที่ต่อหากัน (1/1/5 คือเส้น keepalive)

```
configure terminal
hostname CX-CORE-1-1
interface 1/1/5
  no shutdown
  ip address 192.168.0.2/24
```

2. ที่ CX-CORE-1/2 ทำแบบเดียวกันแต่เปลี่ยนเลข

```
hostname CX-CORE-1-2
interface 1/1/5
  no shutdown
  ip address 192.168.0.3/24
```

3. ที่ CX-CORE-2 ตั้งชื่อไว้ก่อน ส่วน IP 192.168.0.4 ใช้ตอนทำ LAB 4

```
hostname CX-CORE-2
```

### ตรวจสอบผล

```
show interface brief
show ip interface brief
ping 192.168.0.3
```

### ระวัง

ping ไม่ผ่านให้ดูก่อนว่าใส่ IP ผิดตัวหรือเปล่า — console ทั้งสามหน้าต่างเหมือนกันหมดจนกว่าจะตั้ง hostname

## LAB 1 ตั้งชื่อและ IP ให้สวิตช์ทั้งสามตัว (2/2)

PNETLAB (สวิตช์จำลอง) แยกออกว่ากำลังพิมพ์อยู่ที่สวิตช์ตัวไหน และ ping ถึงกันได้ทั้งสามตัว

4. ทดสอบว่าคู่ VSX คุยกันได้  
ping 192.168.0.3
5. บันทึกทุกตัว  
write memory

ตรวจสอบผล

```
show interface brief
show ip interface brief
ping 192.168.0.3
```

ระวัง

ping ไม่ผ่านให้ดูก่อนว่าใส่ IP ผิดตัวหรือเปล่า — console ทั้งสามหน้าต่างเหมือนกันหมดจนกว่าจะตั้ง hostname

## LAB 2 VLAN และพอร์ตที่ต่อ PC (1/2)

PNETLAB (สวิตช์จำลอง) สร้าง VLAN ของผู้ใช้ แล้วเปิดพอร์ตที่ PC ทั้งสามเครื่องเสียบอยู่

1. สร้าง VLAN 10 บนสวิตช์ทั้งสามตัว

```
configure terminal
vlan 10
name USERS
```

2. ที่ CX-CORE-1/1 เปิดพอร์ตที่ PC11 เสียบ

```
interface 1/1/3
no shutdown
no routing
vlan access 10
```

3. ที่ CX-CORE-1/2 เปิดพอร์ตที่ PC12 เสียบ

```
interface 1/1/1
no shutdown
no routing
vlan access 10
```

ตรวจสอบผล

```
show vlan
show interface brief
show mac-address-table
```

ระวัง

ตอนนี้ PC ยังคุยข้ามสวิตช์ไม่ได้ เพราะ VSX กับ LAG ยังไม่ได้ทำ — เป็นเรื่องปกติ ยังไม่ต้องแก้

## LAB 2 VLAN และพอร์ตที่ต่อ PC (2/2)

PNETLAB (สวิตช์จำลอง) สร้าง VLAN ของผู้ใช้ แล้วเปิดพอร์ตที่ PC ทั้งสามเครื่องเสียบอยู่

4. ที่ CX-CORE-2 เปิดพอร์ตที่ PC21 เสียบ

```
interface 1/1/3
  no shutdown
  no routing
  vlan access 10
```

5. ตั้ง IP ให้ PC ทั้งสามเครื่องในวง 192.168.1.0/24 (PC11 = .11, PC12 = .12, PC21 = .21) gateway 192.168.1.1  
แล้วบันทึก config สวิตช์

```
write memory
```

### ตรวจสอบผล

```
show vlan
show interface brief
show mac-address-table
```

### ระวัง

ตอนนี้ PC ยังคุยข้ามสวิตช์ไม่ได้ เพราะ VSX กับ LAG ยังไม่ได้ทำ — เป็นเรื่องปกติ ยังไม่ต้องแก้



## LAB 3 จับคู่ VSX (1/2)

PNETLAB (สวิตช์จำลอง) ทำให้ CX-CORE-1/1 กับ CX-CORE-1/2 เป็นคู่ VSX ที่มองจากข้างล่างเหมือนสวิตช์ตัวเดียว

### 1. สร้าง ISL เป็น LAG บนพอร์ต 1/1/6 — ทำเหมือนกันทั้งสองตัว

```
configure terminal
interface lag 256
    no shutdown
    description VSX-ISL
    no routing
    vlan trunk native 1
    vlan trunk allowed all
    lacp mode active
interface 1/1/6
    no shutdown
    lag 256
```

#### ตรวจสอบผล

```
show vsx status
show vsx brief
show lacp interfaces
```

#### ระวัง

system-mac ต้องเป็นค่าเดียวกันทั้งคู่ ถ้าใส่ไม่ตรงกัน VSX จะไม่ขึ้น · ISL ใช้ 1/1/6 ส่วน keepalive ใช้ 1/1/5 อย่าสลับกัน

## LAB 3 จับคู่ VSX (2/2)

PNETLAB (สวิตช์จำลอง) ทำให้ CX-CORE-1/1 กับ CX-CORE-1/2 เป็นคู่ VSX ที่มองจากข้างล่างเหมือนสวิตช์ตัวเดียว

2. ที่ CX-CORE-1/1 ตั้งเป็น primary — keepalive รังบน 1/1/5 ที่ใส่ IP ไว้ใน LAB 1

```
vsx
system-mac 02:00:00:00:01:00
inter-switch-link lag 256
role primary
keepalive peer 192.168.0.3 source 192.168.0.2
```

3. ที่ CX-CORE-1/2 ตั้งเป็น secondary — system-mac ต้องตรงกัน

```
vsx
system-mac 02:00:00:00:01:00
inter-switch-link lag 256
role secondary
keepalive peer 192.168.0.2 source 192.168.0.3
```

4. บันทึกทั้งสองตัว

```
write memory
```

### ตรวจสอบผล

```
show vsx status
show vsx brief
show lacp interfaces
```

### ระวัง

system-mac ต้องเป็นค่าเดียวกันทั้งคู่ ถ้าใส่ไม่ตรงกัน VSX จะไม่ขึ้น · ISL ใช้ 1/1/6 ส่วน keepalive ใช้ 1/1/5 อย่าสลับกัน

## LAB 4 Active Gateway และ LAG ลง CX-CORE-2 (1/2)

PNETLAB (สวิตช์จำลอง) ให้ PC ทุกเครื่องใช้ gateway เดียวกัน แล้วเชื่อม CX-CORE-2 ขึ้นคู่ VSX ด้วย LAG เส้นเดียว

1. ที่ทั้งสองตัวของคู่ VSX สร้าง SVI ของ VLAN 10 แล้วใส่ active gateway (CX-CORE-1/1 ใช้ .2 · CX-CORE-1/2 ใช้ .3)

```
configure terminal
interface vlan 10
  ip address 192.168.1.2/24
  active-gateway ip 192.168.1.1 mac 02:00:00:00:00:01
```

2. ที่คู่ VSX สร้าง multi-chassis LAG ลง CX-CORE-2 — CX-CORE-1/1 ใช้พอร์ต 1/1/1 · CX-CORE-1/2 ใช้ 1/1/2

```
interface lag 1 multi-chassis
  no shutdown
  no routing
  vlan trunk native 1
  vlan trunk allowed 10
  lacp mode active
interface 1/1/1
  no shutdown
  lag 1
```

### ตรวจสอบ

```
show vsx status
show lacp interfaces
show ip interface brief
show ip route
```

### ระวัง

active-gateway ต้องใส่ทั้งสองตัวและ mac ต้องตรงกัน ไม่งั้น PC จะเห็น gateway หายไปตอน failover

## LAB 4 Active Gateway และ LAG ลง CX-CORE-2 (2/2)

PNETLAB (สวิตช์จำลอง) ให้ PC ทุกเครื่องใช้ gateway เดียวกัน แล้วเชื่อม CX-CORE-2 ขึ้นคู่ VSX ด้วย LAG เส้นเดียว

3. ที่ CX-CORE-2 สร้าง LAG ธรรมดาแล้วใส่ทั้งสองพอร์ตเข้าไป

```
interface lag 1
  no shutdown
  no routing
  vlan trunk native 1
  vlan trunk allowed 10
  lacp mode active
interface 1/1/1-1/1/2
  no shutdown
  lag 1
```

4. ทดสอบ: ping จาก PC21 ไป PC11 และ PC12 ต้องผ่าน และ ping gateway 192.168.1.1 ต้องผ่าน

5. ทดสอบ failover: shutdown พอร์ต 1/1/1 ที่ CX-CORE-1/1 แล้ว ping ต้องยังผ่านอยู่

```
interface 1/1/1
  shutdown
  no shutdown
```

### ตรวจสอบ

```
show vsx status
show lacp interfaces
show ip interface brief
show ip route
```

### ระวัง

active-gateway ต้องใส่ทั้งสองตัวและ mac ต้องตรงกัน ไม่งั้น PC จะเห็น gateway หายไปตอน failover

## DEMO Remote AP และสวิตช์จริง — ผู้สอนสาธิต

หน้าห้อง เห็นว่าอุปกรณ์จริงที่ทุกชุดใช้ร่วมกันต่อกันอย่างไร และทำไมนักศึกษาไม่ได้แตะ

1. ดู AP-515 ที่ตั้งเป็น Remote AP — จบ IPsec tunnel ที่ 9004-LTE ดึงวงของแลบมาถึงห้องนี้
2. ดูสวิตช์ CX 6100 — พอร์ต 1/1/12 เป็น uplink ขึ้น RAP ส่วนพอร์ต 1/1/1–1/1/8 คือ Campus AP ของแต่ละชุด
3. เปิด configs/SWITCH.txt ดูว่า VLAN ของแต่ละชุดถูกวางไว้อย่างไร
4. ดู PoE budget ว่าเหลือเท่าไรเมื่อเสียบ AP ครบทุกชุด  
show power-over-ethernet brief

### ตรวจสอบ

```
show ap database
show vlan
show power-over-ethernet brief
```

### ระวัง

สามตัวนี้เป็นทางผ่านของทุกชุด ถ้าลัมคือลัมทั้งห้อง — จึงเป็นผู้สอน ส่วนที่ฝึกพิมพ์อยู่บน PNETLAB



# ArubaOS-CX switch CLI contexts

## Operator (>)

- Enables execution of commands to view—but not change—the configuration
- Switch prompt:
  - switch>

## Manager (#)

- From manager context (#), execute commands that do not require saving changes to configuration
- Switch prompt:
  - switch#

## Global configuration (config)#

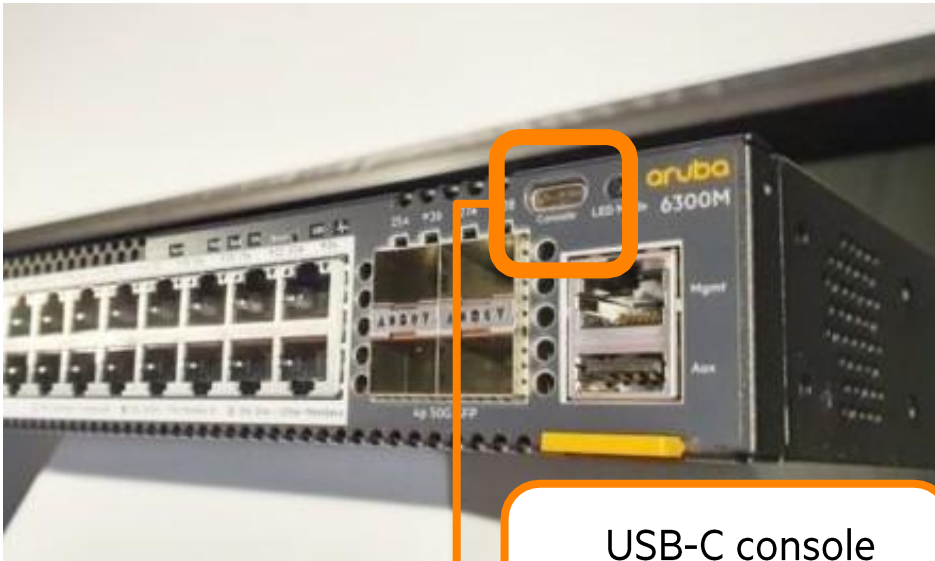
- From global configuration context (config), execute commands that change configuration of switch
- Switch prompt:
  - switch(config)#

## Other configuration

- switch(config-if)#
- switch(config-router)#
- switch(config-vlan-100)#

Note: If you log into a device with the admin role you are immediately put into enable context. Type 'exit' at enable to disconnect (not be put into operator mode).

# Connecting to Console Port



USB-C console  
No RJ45 console  
(CX6300)



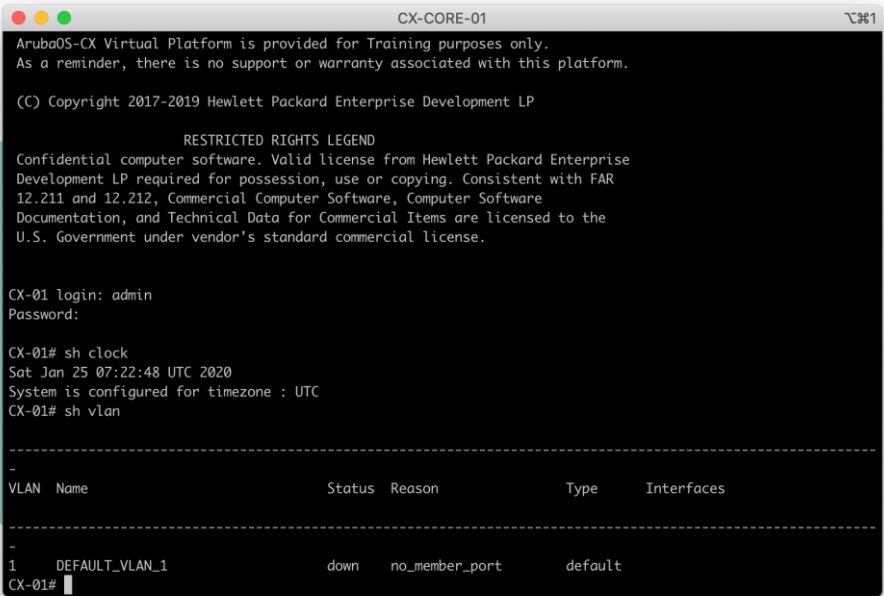
CX Switch

Direct serial connection to console port

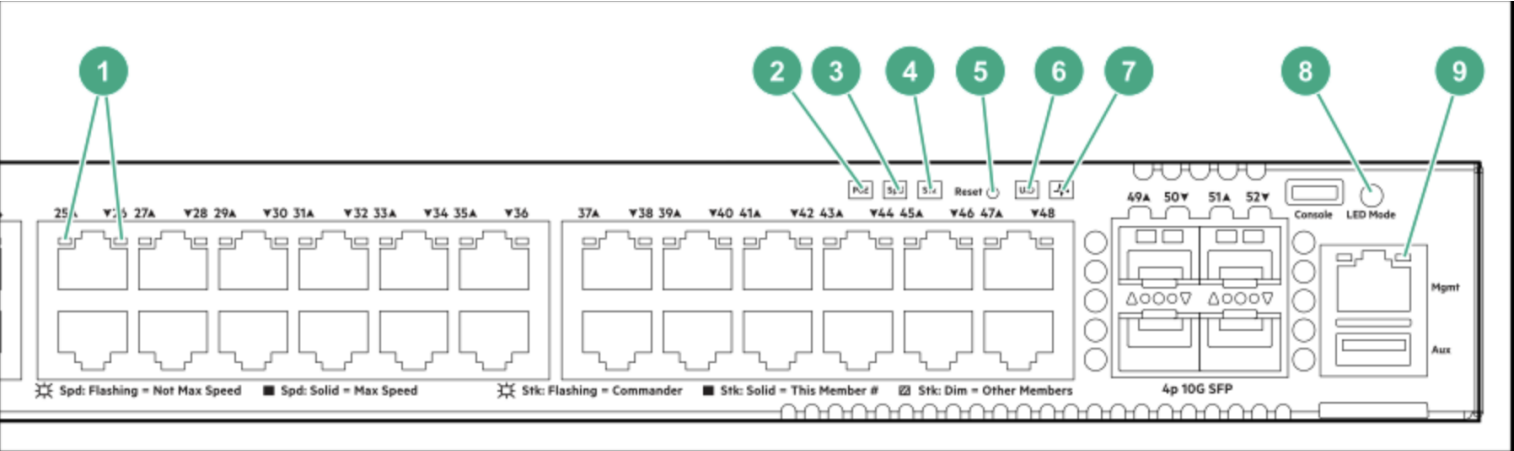
| CX Switch        | Serial Console Type  |
|------------------|----------------------|
| 8400, 8325-48Y8C | RJ45 and Micro USB-B |
| 8320, 8325-32C   | RJ45                 |
| 6400             | RJ45 and USB-C       |
| 6300             | USB-C                |
| 6200 (12 port)   | USB-C, RJ45          |

### Terminal software settings

- Speed: 115200 bps
- Data bits: 8
- Stop bits: 1
- Parity: None
- Flow control: None



# Switch LEDs



| Label | Description                |
|-------|----------------------------|
| 1     | Switch port LEDs           |
| 2     | PoE LED (only support POE) |
| 3     | Speed mode selected LED    |
| 4     | Stk LED                    |
| 5     | Reset button               |
| 6     | UID (Unit Identification)  |
| 7     | Global Status LED          |
| 8     | Mode select button         |
| 9     | OOBM port LED              |

# User Accounts

- admin is the default user with **no password**
- Administrators
  - Users with administrator rights can execute any command.
  - Default setting will have admin user account without password, and you are required to enter a **new password after login**
- Operators
  - Users with operator rights can execute commands in the operator context (>) only
- Auditors
  - Users with auditor rights can execute commands in the auditor context (**auditor>**) only
  - Limited view of show command

# Lab 1.1 – Initialize Admin Account and Change Hostname

```
switch login: admin  
Password:
```

```
Please configure the 'admin' user account  
password. Enter new Password: ****  
Confirm new password: ****
```

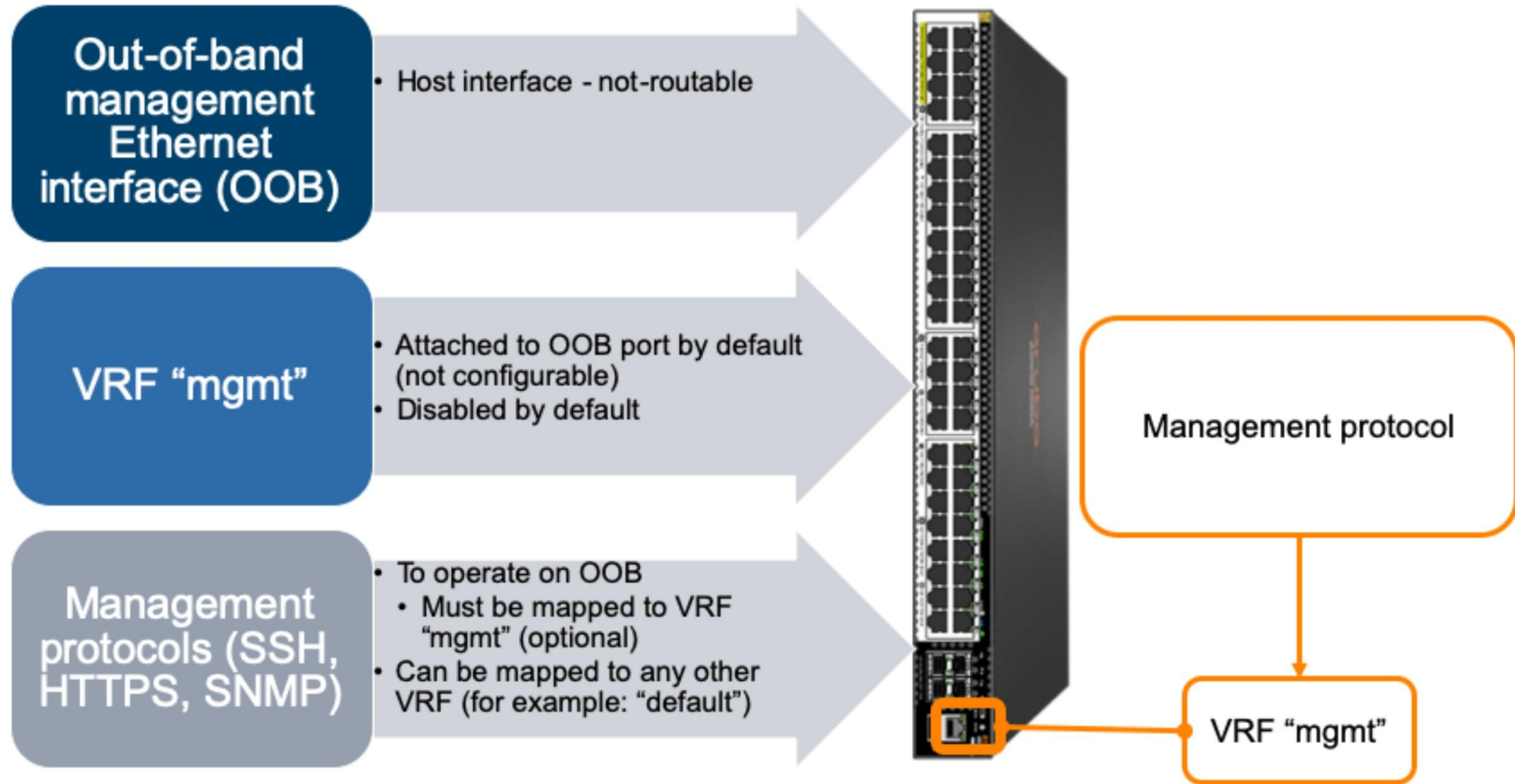
1. Default setting will have admin user account without password, and you are required to enter a new password after login
2. Enter any password string you want, we will change to the new one in the next lab

```
Switch# config terminal  
Switch(config)# hostname CX-CORE-01  
CX-CORE-01(config)#
```

1. Change switch name to CX-CORE-01

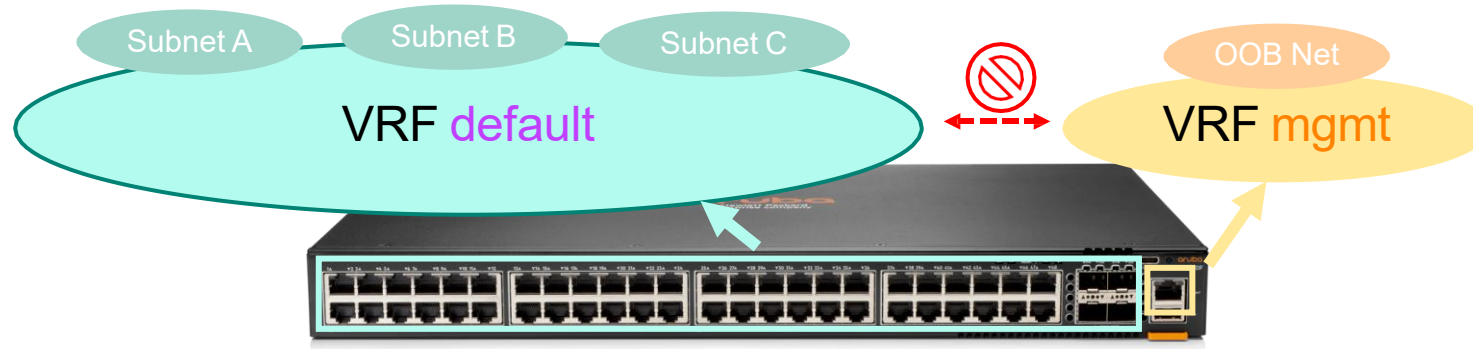


# Out of band Management



# VRF-lite in ArubaOS-CX

- VRF technology is an integral part of the operating system
- There is two default VRF called default and mgmt
  - The management interface belongs to the **mgmt**
  - All data L3 interfaces belong to the **default**



have a password

- Must be attached to a VRF

- “mgmt” for OOB management
- “default”
- Any other

for multiple VRFs

supported



```

15.529.82.105 - Tera Term VT
File Edit Setup Control Window Help

lag 128
interface 1/1/32
no shutdown
lag 128
interface lagback 1
no shutdown
ip vrf 1 area 0.0.0.0
interface lagback 50
no shutdown
vrf attach GRE
ip address 192.168.2.2/32
interface tunnel 1 made gre ip=4
source ip 1.1.2.2
destination ip 1.1.2.1
ip address 192.168.1.2/30
no shutdown
vrf attach GRE
interface vlan30
ip address 10.0.30.1/24
ip vrf 1 area 0.0.0.0
ip route 192.168.2.1/32 192.168.1.1 vrf GRE
http-server vrf access-mode read-write
http-server vrf ngnat
n400-UPP08

```

VRF "mgmt"

# HTTPS Server - WebUI

Requires the admin user to have a password

Must be attached to a VRF

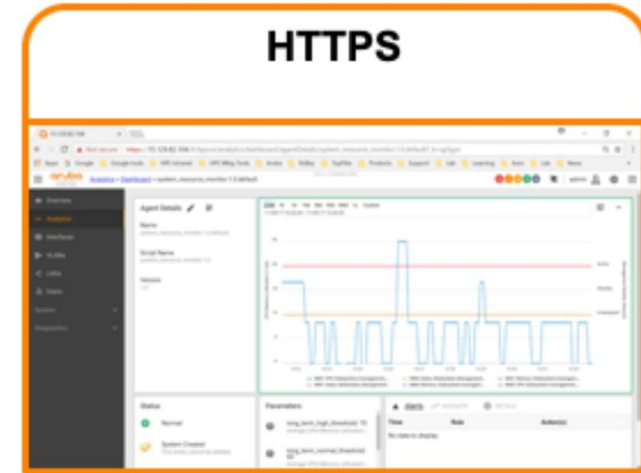
- “mgmt” for OOB management
- “default”
- Any other

Can be assigned simultaneously to multiple VRFs

API is accessible in read-only or read-write

Web-UI is disabled by default

Note: HTTP server is not supported



VRF “mgmt”

# REST Interface

Requires admin user to have a password

REST reached via HTTPS

Access can be read-write or read-only (default read-only)



## REST

ArubaOS-CX REST API

RESTful interface for ArubaOS-CX switch software

|                            |           |                 |                   |
|----------------------------|-----------|-----------------|-------------------|
| AAA_Server_Group           | Show/Hide | Lock Operations | Expand Operations |
| AAA_Server_Group_Profile   | Show/Hide | Lock Operations | Expand Operations |
| ACL                        | Show/Hide | Lock Operations | Expand Operations |
| ACL_Entry                  | Show/Hide | Lock Operations | Expand Operations |
| Aggregate_address          | Show/Hide | Lock Operations | Expand Operations |
| SGP_ASPub_Filter           | Show/Hide | Lock Operations | Expand Operations |
| SGP_ASPub_Filter_Entry     | Show/Hide | Lock Operations | Expand Operations |
| SGP_Community_Filter       | Show/Hide | Lock Operations | Expand Operations |
| SGP_Community_Filter_Entry | Show/Hide | Lock Operations | Expand Operations |
| SGP_Neighbor               | Show/Hide | Lock Operations | Expand Operations |
| SGP_Route                  | Show/Hide | Lock Operations | Expand Operations |
| SGP_Router                 | Show/Hide | Lock Operations | Expand Operations |

VRF "mgmt"



# Lab 1.3 - Enable Remote Management Access

```
CX-CORE-01# configure terminal
CX-CORE-01(config)# https-server vrf mgmt
CX-CORE-01(config)# ssh server vrf mgmt
CX-CORE-01(config)# https-server rest access-mode read-only
CX-CORE-01(config)# interface mgmt
CX-CORE-01(config-if-mgmt)# ip static 10.1.10.101-103/24
CX-CORE-01(config-if-mgmt)# default-gateway 10.1.10.1
CX-CORE-01(config-if-mgmt)# nameserver 8.8.8.8
CX-CORE-01(config-if-mgmt)# no shutdown
```

- 1. Bind the https and ssh server with VRF “mgmt.”
- 2. Assign IP address, Gateway and DNS to the management interface

- 1. Open the browser and see if you can access CX switches
- 2. SSH CX switches and make sure you have the access
- 3. Ping to the switch IP address on mgmt VRF

**MGMT IP Address**  
Group1 - CX-CORE = 10.1.10.101-103/24  
Group2 - CX-CORE = 10.1.10.104-106/24  
Group3 - CX-CORE = 10.1.10.107-109/24  
Group4 - CX-CORE = 10.1.10.110-112/24  
Group5 - CX-CORE = 10.1.10.113-115/24  
Group6 - CX-CORE = 10.1.10.116-118/24  
Group7 - CX-CORE = 10.1.10.119-121/24

ping 10.1.10.101  
PING 10.1.10.101 (10.1.10.101) 100(128) bytes of data.

|     |       |      |              |            |         |               |
|-----|-------|------|--------------|------------|---------|---------------|
| 108 | bytes | from | 10.1.10.101: | icmp_seq=1 | ttl=128 | time=1.04 ms  |
| 108 | bytes | from | 10.1.10.101: | icmp_seq=2 | ttl=128 | time=0.454 ms |
| 108 | bytes | from | 10.1.10.101: | icmp_seq=3 | ttl=128 | time=0.763 ms |
| 108 | bytes | from | 10.1.10.101: | icmp_seq=4 | ttl=128 | time=0.681 ms |
| 108 | bytes | from | 10.1.10.101: | icmp_seq=5 | ttl=128 | time=0.604 ms |

# Lab 1.4 – Configure Clock, Timezone, DNS and SNMP

- Some of the basic configuration like DNS and Logging supports VRF binding

```
CX-CORE-01 # configure terminal
CX-CORE-01(config)# clock timezone Asia/Bangkok
CX-CORE-01(config)# ip dns server-address 8.8.8.8 vrf mgmt
CX-CORE-01(config)# ntp server 0.th.pool.ntp.org
CX-CORE-01(config)# ntp vrf mgmt
CX-CORE-01(config)# ntp enable
CX-CORE-01(config)# snmp-server system-description CX-CORE-01
CX-CORE-01(config)# snmp-server system-contact Administrator
CX-CORE-01(config)# snmp-server system-location CX_LAB
CX-CORE-01(config)# snmp-server community cx_snmp
CX-CORE-01(config)# snmp-server vrf mgmt
CX-CORE-01(config)# snmp-server host x.x.x.x trap version v2c community cx_snmp
CX-CORE-01(config)# logging x.x.x.x vrf mgmt
```

- SNMP trap and log server are dummy IP address. It is intended for student to practice how to config these two settings.

```
CX-CORE-01# sh ntp associations
```

| ID  | NAME              | REMOTE       | REF-ID | ST | LAST | POLL | REACH |
|-----|-------------------|--------------|--------|----|------|------|-------|
| * 1 | 0.th.pool.ntp.org | 202.12.97.45 | .PPS.  | 1  | 7    | 64   | 377   |

```
CX-CORE-01# show clock
Tue Jan 28 13:50:39 ICT 2020
System is configured for timezone : Asia/Bangkok
```

- Verify date, time and timezone

# Configuration Management

- “**write memory**” to save running configuration into the Startup configuration
- **Types of checkpoints**
  - **System generated checkpoints:** Checkpoints that the system automatically generates whenever a configuration change is detected.
  - **User generated checkpoints:** Checkpoints that the user manually generates
- Total limit of checkpoint is **64** including the startup configuration; **32** each type
- After the limit the system will overwrite the oldest checkpoint
- For each additional configuration change, the **300** second timeout counter is reset. Once the timeout limit is reached with no additional configuration changes detected, the system will automatically generate a new checkpoint
- This system generated checkpoints are listed with a time stamp format CPC<**YYYYMMDDHHMMSS**>
- The checkpoints can be applied using the “**checkpoint rollback**” or “**copy checkpoint**” command
- **Difference** between two configuration can be viewed between two different checkpoints or between any checkpoint and the startup/running configuration

## Lab 1.5 – Creating Configuration Checkpoint

```
CX-CORE-01 # copy running-config checkpoint initial_config  
Large configuration changes will take time to process, please be patient.
```

```
CX-CORE-01# show checkpoint  
CPC20200128072408  
CPC20200128075706  
initial_config
```

- Use this command to show all saved configuration checkpoint

```
CX-CORE-01# show checkpoint initial_config  
.  
.  
.
```

- Use this command to show initial\_config checkpoint

# Lab 1.7 – Checkpoint Rollback

Use the **same CX switch in lab 1.6** to do this lab

```
CX-CORE-01# sh session-timeout
session-timeout: 0 minute
```

```
CX-CORE-01# sh banner motd
#####
TEST BANNER
#####
```

```
CX-CORE-01# sh snmp community
-----
SNMP communities
-----
public
```

1. Record the show output result and verify the difference after perform a configuration rollback

2. After that, issue a rollback command. You have two way to rollback your configuration to any checkpoint  
3. Verify the show out result after rollback your configuration. What is the difference?

```
CX-CORE-01# copy checkpoint initial_config running-config
Configuration changes will take time to process, please be patient.
```

Option 1

```
CX-CORE-01# checkpoint rollback initial_config
Configuration changes will take time to process, please be patient.
```

Option 2

## Knowledge Check #1

**Which console interfaces and default baud rates are supported on the HPE Aruba CX 6200F 12-port model? (Select two.)**

- a. USB-A, baud rate = 115200
- b. USB-C, baud rate = 9600
- c. RJ-45, baud rate = 115200
- d. USB-C, baud rate = 115200
- e. RJ-45, baud rate = 112500



## Knowledge Check #2

---

**Does the HPE Aruba 6200F model support only two VRF?**

- a. True.
- b. False.

## Knowledge Check #3

---

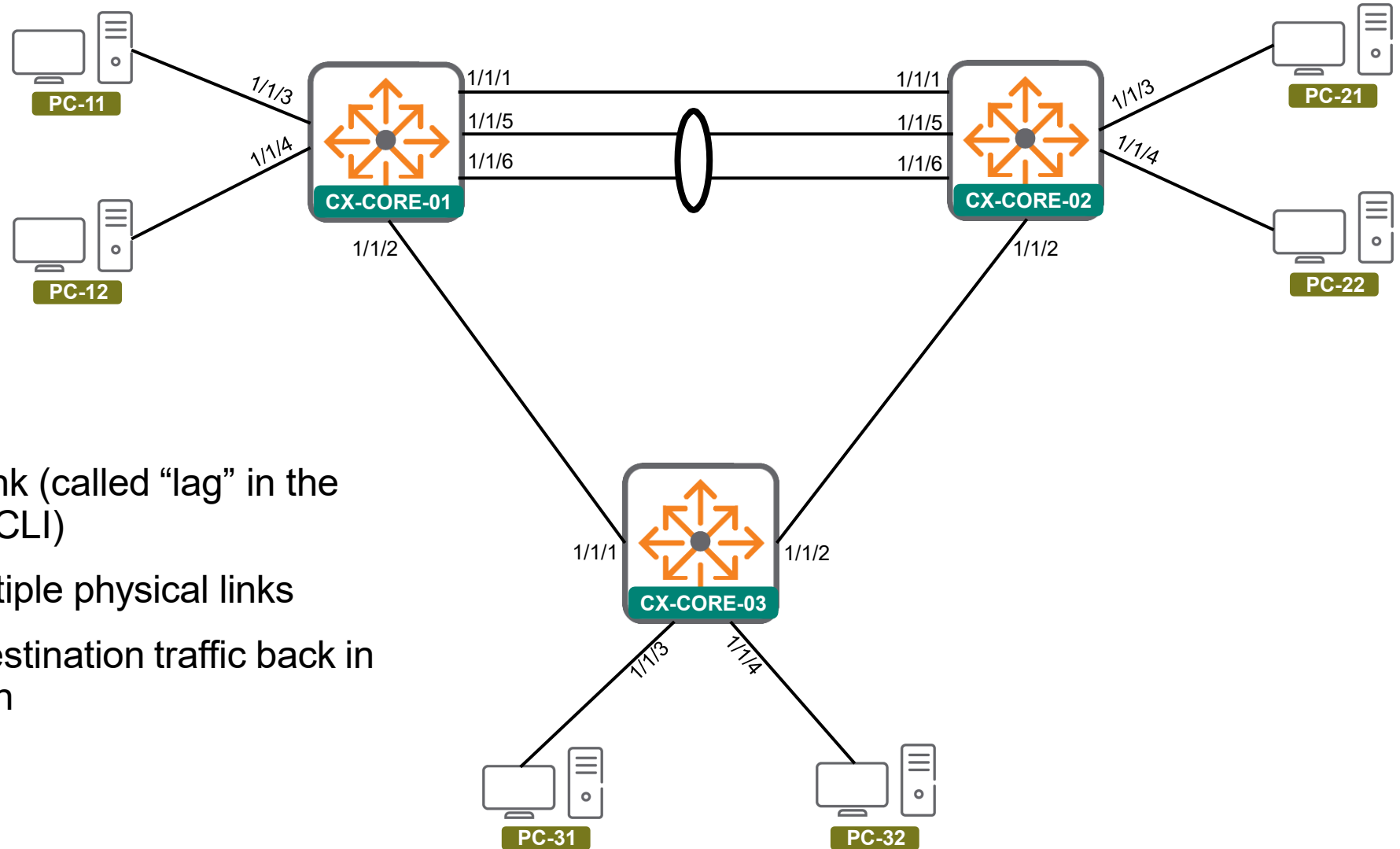
**The REST API on Aruba CX supports both read and write operations (GET, POST, PUT, DELETE)?**

- a. True.
- b. False.

# Lab 2

# Layer 2 Switching and Features

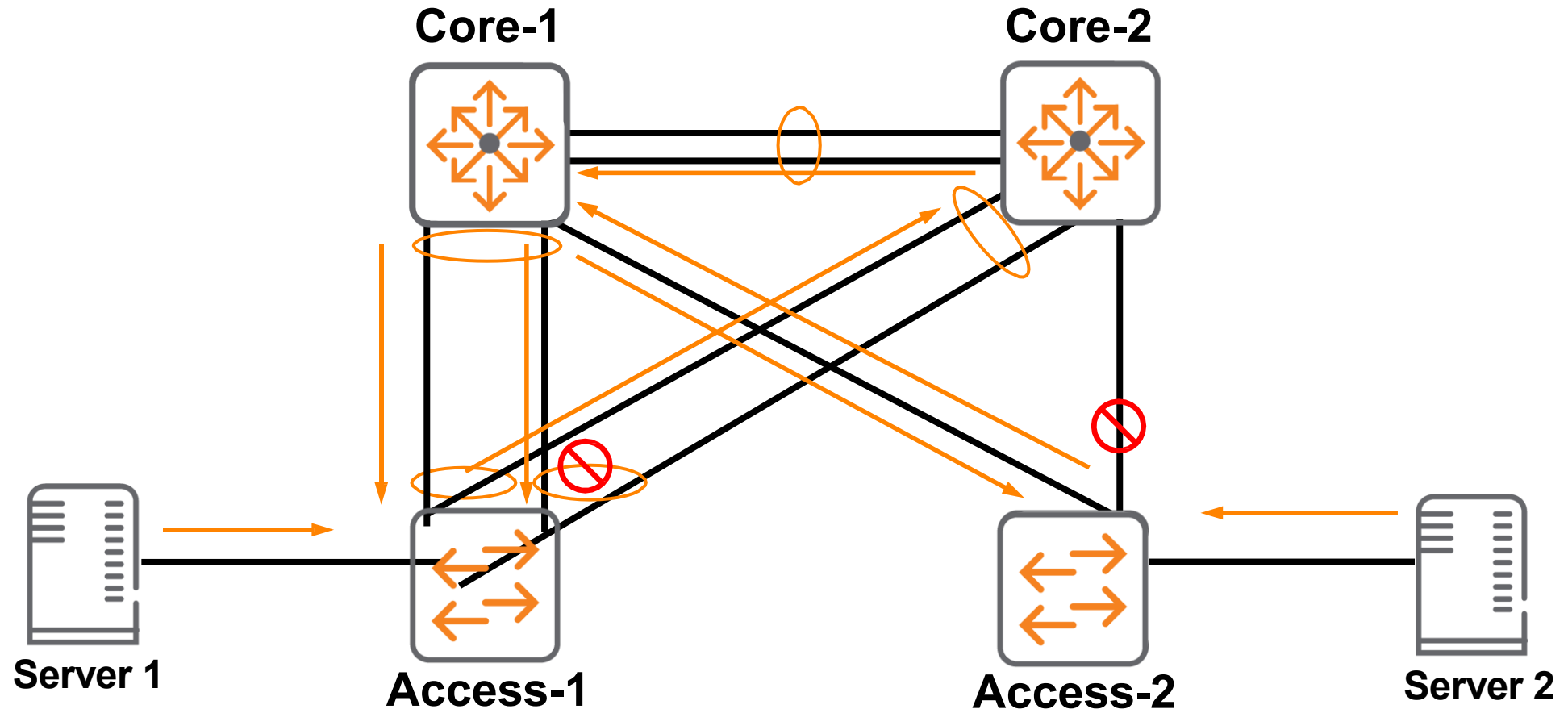
# Link Aggregation



- Acts as one logical link (called “lag” in the ArubaOS-CX switch CLI)
- Carries traffic on multiple physical links
- Never floods multi-destination traffic back in the same aggregation

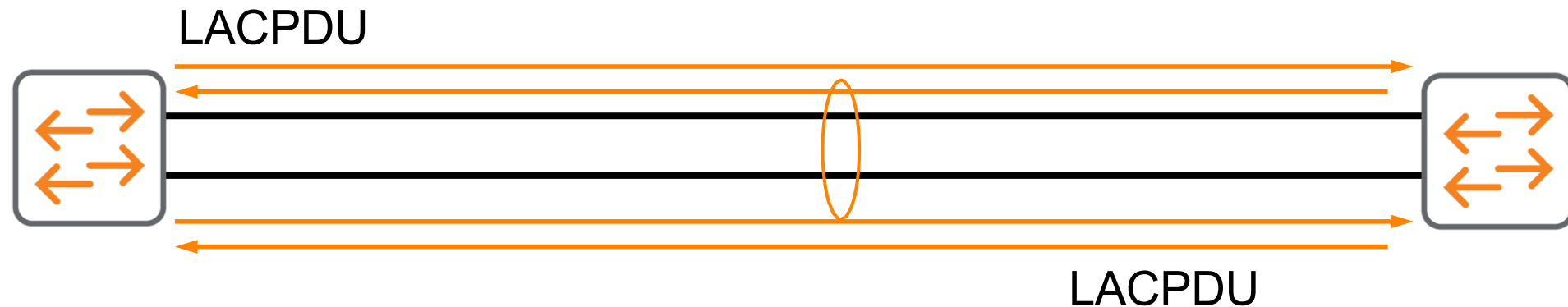
## Potential issue with manual link aggregation

- What if you accidentally configure the wrong ports in a link aggregation?



# LACP

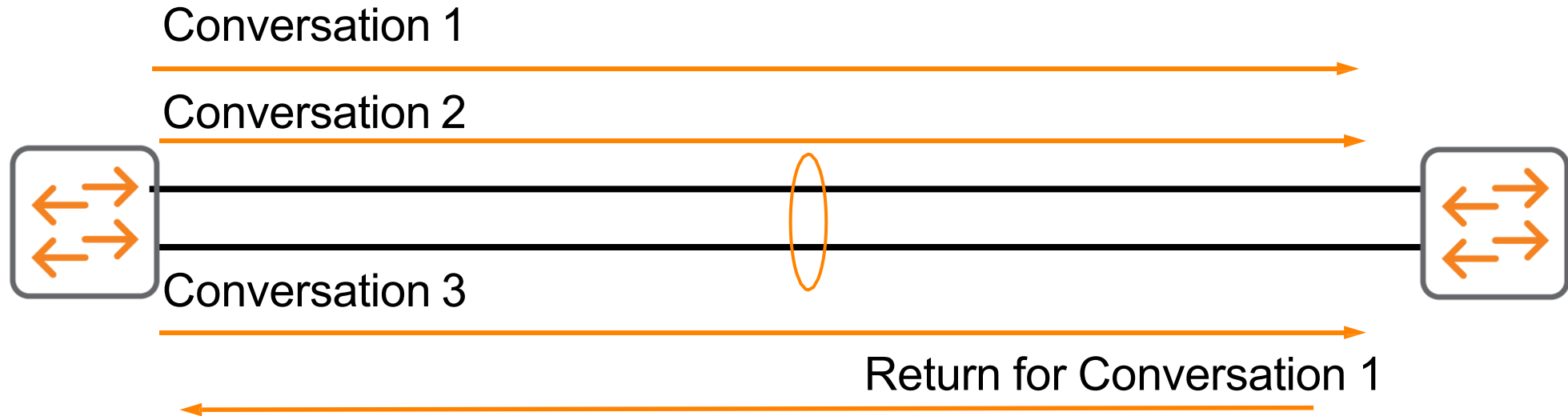
- Uses messages to establish the link aggregation
- Ensures that all links are connected to the same peer (system ID) and link aggregation (operational key)
- Ensures compatibility for other settings
- Manages adding and removing links



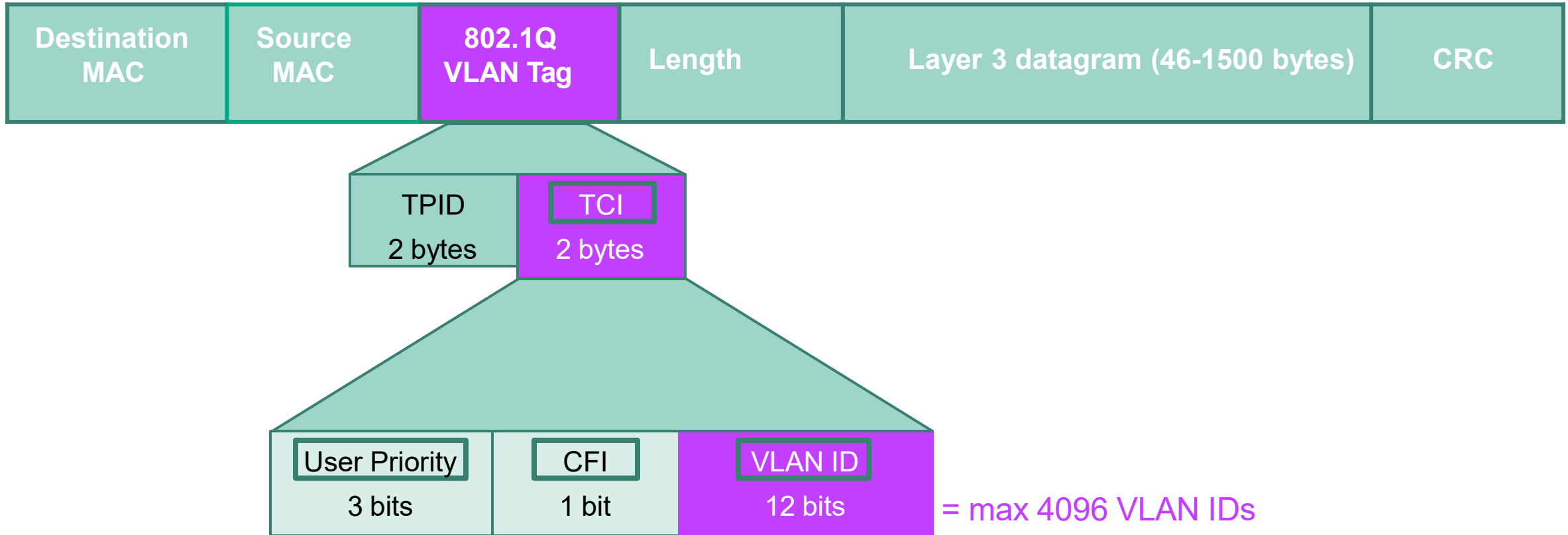


# Load-sharing traffic over a link aggregation

- The link aggregation uses a hash to select a link for each *conversation*:
  - Conversation defined by the load-sharing mode
    - MAC source and destination address (L2-based)
    - IP source and destination address (L3-based) = Default option
    - Source and destination UDP/TCP ports (L4-based) = Only 8400 series switches support L4 hashing
  - Link for a conversation stays constant
  - Traffic tends to be balanced more evenly as conversations increase



# Ethernet Frame Structure



## Knowledge Check

---

**What feature allows combining multiple physical links into a single logical interface?**

- a. VLAN
- b. STP
- c. LACP (Link Aggregation Control Protocol)
- d. LLDP



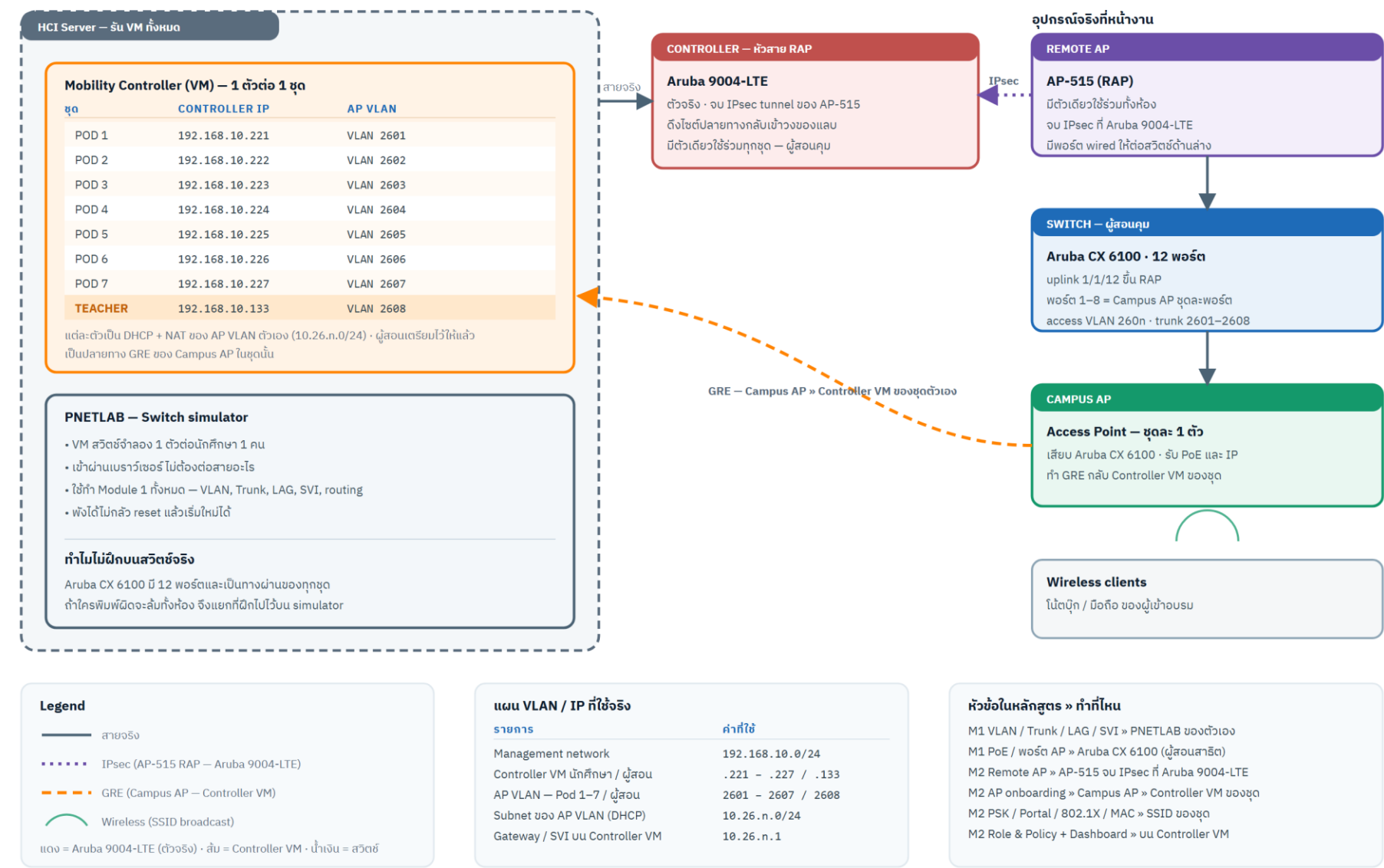
# WLAN Labs

---



# Aruba Campus Network Training — Lab Topology

RMUTT x Commserv Siam · 26 ส.ค. 2569 · RAP จบที่ Aruba 9004-LTE · Campus AP จบที่ Controller VM ของชุดตัวเอง · 7 ชุดนักศึกษา + 1 ชุดผู้สอน



## LAB 5 เข้า Controller VM และอ่านค่าที่เตรียมไว้

Controller VM ของชุดตัวเอง เข้าใช้ controller ของชุดตัวเองได้ และรู้ว่าอะไรถูกตั้งไว้ให้แล้วบ้าง

1. เข้าเว็บ controller ของชุดตัวเอง (POD 1 = <https://192.168.10.221:4343>) ผู้ใช้ admin
2. ดู license AP และ PEF ที่ Configuration → License — ผู้สอนใส่ไว้แล้ว
3. ดูเวลาและ timezone ที่ Configuration → System → General ต้องเป็น Asia/Bangkok
4. ดู country code ต้องเป็น TH ไม่งั้น AP จะไม่ยอมกระจายสัญญาณ

### ตรวจสอบ

```
show switchinfo  
show license  
show clock
```

### ระวัง

เวลาผิดพลาดให้ certificate ของ AP ไม่ผ่านแล้ว AP จะไม่ขึ้น — ถ้าเจอ AP ไม่ขึ้นทั้งชุด ให้ดูเวลาก่อน



## LAB 6 VLAN, SVI, DHCP และ NAT บน Controller

Controller VM ของชุดตัวเอง ทำให้ controller เป็น gateway และ DHCP server ของ AP VLAN ของตัวเอง แล้ว NAT ออกเน็ต

1. เปิดหัวข้อ Config ด้านบน เลือกแท็บชุดตัวเอง กดคัดลอก แล้ววางลง CLI ของ controller
2. อ่านทีละท่อนว่าแต่ละบล็อกทำอะไร — VLAN กับ SVI ที่มี ip nat inside → ช่วง IP ที่กันไม่ให้แจก → DHCP pool → service dhcp
3. บันทึก  
write memory

### ตรวจสอบ

```
show ip interface brief
show ip dhcp binding
show ip dhcp statistics
```

### ระวัง

service dhcp ไม่เปิด pool จะไม่ทำงานแม้ config ถูกหมด ·  
เพลอวางของชุดอื่นให้ลบด้วย no interface vlan และ no ip  
dhcp pool ก่อนวางใหม่

## LAB 7 Campus AP onboarding

Controller VM ของชุดตัวเอง + AP เอา Campus AP ของชุดตัวเอง (1 ตัว) ขึ้นมาอยู่กับ controller ของชุด และแยกเป็น AP group

1. AP ของชุดตัวเองเสียบอยู่ที่ CX 6100 พอร์ตของชุดนั้นแล้ว (POD 1 = 1/1/1) — รับ PoE และ IP จาก DHCP บน controller
2. ถ้า AP เคยใช้ที่อื่นมาก่อน ให้ reset โดยกดปุ่ม reset ค้างเกิน 10 วินาทีตอนเปิดเครื่อง
3. ชี้ AP มาที่ controller ของชุดตัวเอง (convert to campus mode)
4. อนุมัติ AP ที่ Configuration → Access Points → Allowlist → Approve
5. สร้าง AP group ของชุดตัวเอง แล้วย้าย AP เข้ากลุ่ม

### ตรวจสอบ

```
show ap database  
show ap active  
show ip dhcp binding
```

### ระวัง

AP ไม่ขึ้นให้ไล่จากล่างขึ้นบน — พอร์ตบน CX 6100 ไขของชุดเราใหม่ → ได้ IP ใหม่ → เห็นใน show ap database ใหม่

## LAB 8 WLAN — PSK และ MAC Authentication

Controller VM ของชุดตัวเอง สร้าง SSID แรกด้วย PSK แล้วเพิ่มการตรวจ MAC จากฐานข้อมูลในตัว controller

1. สร้าง WLAN ที่ Configuration → WLANs → + ตั้งชื่อ SSID เป็นของชุดตัวเอง เช่น POD1-IOT
2. เลือก AP group ของตัวเอง · VLAN 2601 · Forwarding mode Tunnel
3. ตั้ง security เป็น Personal (PSK) แล้วทดสอบด้วยโน้ตบุ๊ก
4. เพิ่ม MAC ของเครื่องทดสอบเป็น local user ที่ Configuration → Authentication → Internal Server → Users โดยใช้ MAC เป็นทั้ง username และ password
5. เปิด MAC authentication บน WLAN เดิม แล้วทดสอบใหม่

### ตรวจสอบผล

```
show user-table  
show ap association
```

### ระวัง

ใส่ MAC ผิดฟอร์แมตคือปัญหาที่เจอบ่อย — ดู MAC จริงจาก show user-table ของเครื่องที่เพิ่งต่อ แล้วคัดลอกไปใส่

## LAB 9 WLAN — Captive portal สำหรับ Guest

Controller VM ของชุดตัวเอง สร้าง SSID สำหรับผู้มาเยือน ที่ต้องล็อกอินผ่านหน้าเว็บก่อนใช้งาน

1. สร้าง WLAN ใหม่ ชื่อ POD1-GUEST · VLAN 2601 · AP group ของตัวเอง
2. เลือก security เป็น Captive portal (Guest)
3. สร้าง guest user ใน Internal Server
4. ต่อด้วยมือถือ — ต้องหน้าจอ login ก่อนถึงจะออกเน็ตได้

ตรวจสอบผล

```
show user-table  
show rights
```

ระวัง

ไม่หน้าจอ login ให้ดูว่าเครื่องได้ IP กับ DNS จริงหรือยัง —  
captive portal ต้องให้ DNS ผ่านก่อน

## LAB 10 WLAN — 802.1X ด้วย Internal Server

Controller VM ของชุดตัวเอง สร้าง SSID ที่ผู้ใช้งานใส่ username และ password ของตัวเอง โดยตรงกับฐานข้อมูลในตัว controller

1. สร้าง WLAN ใหม่ ชื่อ POD1-EMP · VLAN 2601 · AP group ของตัวเอง
2. เลือก security เป็น Enterprise (802.1X) แล้วเลือก auth server เป็น Internal
3. สร้าง user ที่ Configuration → Authentication → Internal Server → Users
4. ต่อด้วยโน้ตบุ๊ก ใส่ username และ password ที่สร้างไว้

### ตรวจสอบผล

```
show user-table  
show aaa authentication-server internal
```

### ระวัง

งานนี้ไม่มี ClearPass — 802.1X ทำได้เฉพาะฝั่ง WLAN ที่ controller ตรวจสอบเอง ฝั่ง wired ไม่มี RADIUS ให้ชี้

## LAB 11 Role, Policy และ Dashboard

Controller VM ของชุดตัวเอง จำกัดสิทธิ์ผู้ใช้ด้วย role แล้วดูผลบน dashboard

1. สร้าง network alias ของวงที่จะห้าม ที่ Configuration → Roles & Policies → Aliases
2. สร้าง policy ที่ deny ปลายทางเป็น alias นั้น แล้ว permit ที่เหลือ
3. สร้าง role ใหม่ แล้วผูก policy เข้าไป
4. ผูก role ให้ user ที่สร้างไว้ใน LAB 10 แล้วต่อใหม่
5. ดูผลที่ Dashboard → Clients และดู AppRF ว่าเห็น traffic อะไรบ้าง

### ตรวจสอบผล

```
show user-table  
show rights  
show datapath session table
```

### ระวัง

role ที่ผูกกับ user มีผลตอนต่อใหม่เท่านั้น — ตัดการเชื่อมต่อแล้วต่อใหม่ก่อนสรุปว่า policy ไม่ทำงาน

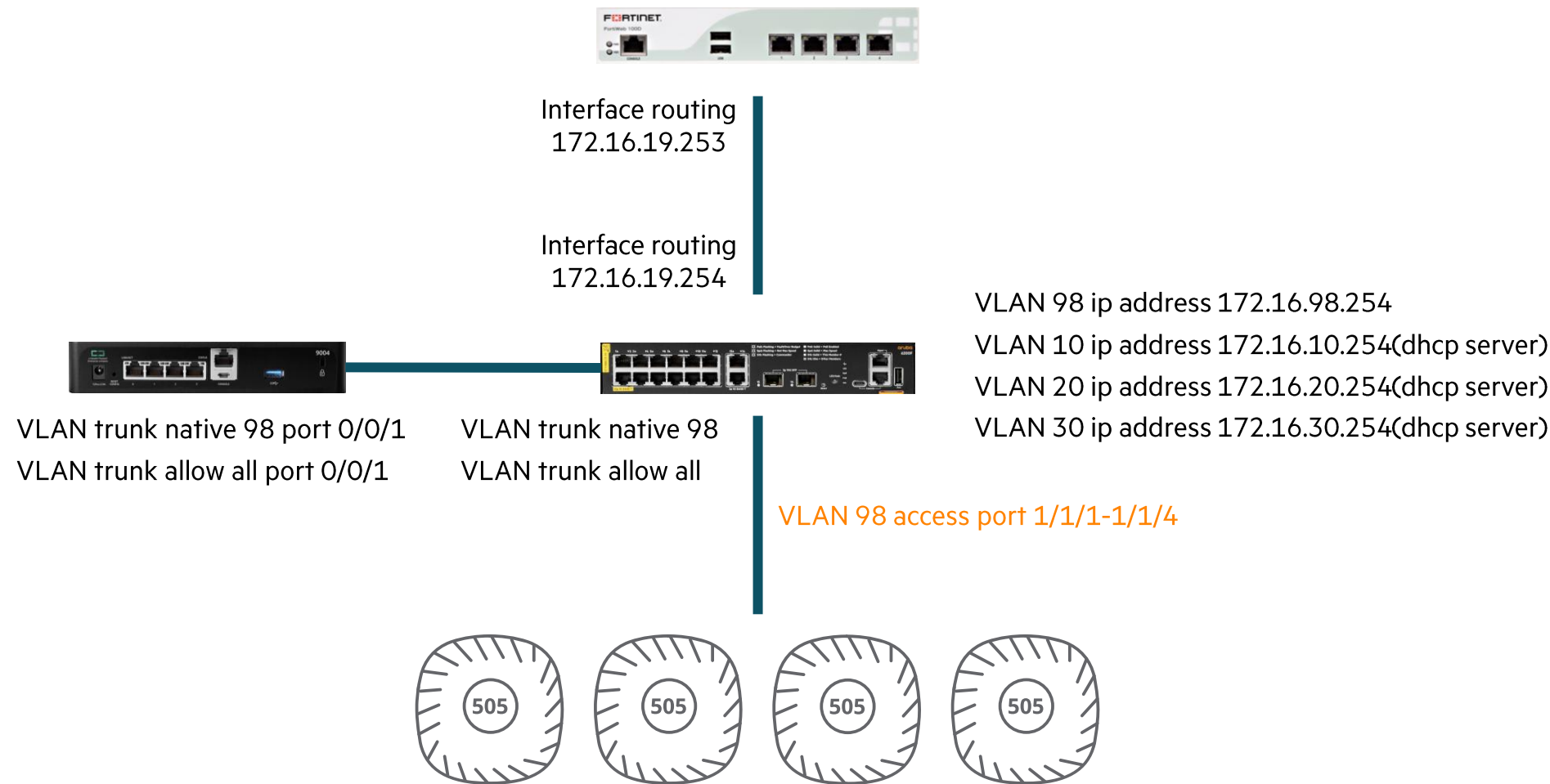


# A. Diagram

---



# Diagram





# B.Console Cable

---



# C.Initial Mobility Controller(MC)

---

- **Initial Controller**
- **Generate trail license AP & PEF**
- **Add License AP & PEF**
- **Set Time**
- **Configure CPSec**



# Enter initial command : Full-setup

```
Auto-provisioning is in progress. It requires DHCP and Activate servers
Choose one of the following options to override or debug auto-provisioning...
'enable-debug'      : Enable auto-provisioning debug logs
'disable-debug'     : Disable auto-provisioning debug logs
'mini-setup'        : Start mini setup dialog. Provides minimal customization and requires DHCP server
'full-setup'        : Start full setup dialog. Provides full customization
'static-activate'   : Provides customization for static or PPPoE ip assignment. Uses activate for conductor information

Enter Option (partial string is acceptable): full-setup

Are you sure that you want to stop auto-provisioning and start full setup dialog? (yes/no): yes
```

Auto-provisioning is in progress. It requires DHCP and Activate servers  
Choose one of the following options to override or debug auto-provisioning....

- 'enable-debug' : Enable auto-provisioning debug logs
- 'disable-debug' : Disable auto-provisioning debug logs
- 'mini-setup' : Start mini setup dialog. Provides minimal customization and requires DHCP server
- ' **full - setup** ' : Start full setup dialog. Provides full customization
- 'static-activate' : Provides customization for static or PPPoE ip assignment. Uses activate for conductor information

Enter Option (partial string is acceptable): **full-setup**

Are you sure that you want to stop auto-provisioning and start full setup dialog? (yes/no): **yes**

# Initial MC-0X

Commands: <Enter> Submit input or use [default value], <ctrl-I> Help  
<ctrl-B> Back, <ctrl-F> Forward, <ctrl-A> Line begin, <ctrl-E> Line end  
<ctrl-D> Delete, <BackSpace> Delete back, <ctrl-K> Delete to end of line  
<ctrl-P> Previous question <ctrl-X> Restart beginning <ctrl-R> Reload box

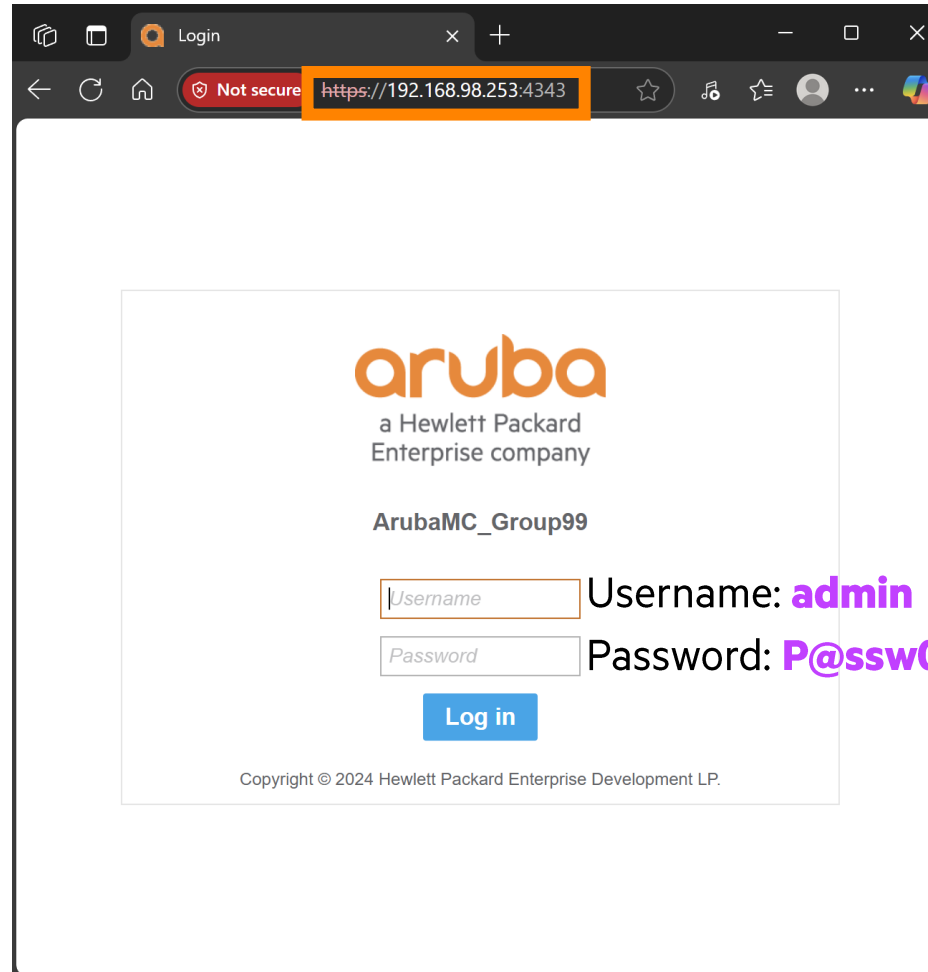
```
Enter System name [Aruba7210_01_B1_F8]: ArubaMC_Group99
Enter Switch Role (conductor|standalone|md) [md]: standalone
Enter Controller VLAN ID [1]: 98
Enter Controller VLAN port [GE 0/0/0]:
Enter Controller VLAN port mode (access|trunk) [access]: trunk
Enter Native VLAN ID [1]: 98
Do you wish to configure IPV4 address on vlan (yes|no) [yes]: yes
Enter VLAN interface IP address [172.16.0.254]: 192.168.98.253
Enter VLAN interface subnet mask [255.255.255.0]: 255.255.255.0
Enter IP Default gateway [none]: 192.168.98.254
Enter DNS IP address [none]: 8.8.8.8
Do you wish to configure IPV6 address on vlan (yes|no) [yes]: no
Enter Country code (ISO-3166), <ctrl-I> for supported list: TH
You have chosen Country code TH for Thailand (yes|no)?: yes
Enter the controller's IANA Time zone [America/Los_Angeles]: Asia/Bangkok
Enter Time in UTC [07:01:44]: 14:04:00
Enter Date (MM/DD/YYYY) [6/14/2025]:
Enter Password for admin login (up to 32 chars): *****
Re-type Password for admin login: *****
```

Commands: <Enter> Submit input or use [default value], <ctrl-I> Help  
<ctrl-B> Back, <ctrl-F> Forward, <ctrl-A> Line begin, <ctrl-E> Line end  
<ctrl-D> Delete, <BackSpace> Delete back, <ctrl-K> Delete to end of line  
<ctrl-P> Previous question <ctrl-X> Restart beginning <ctrl-R> Reload box

```
Enter System name [Aruba7210_01_B1_F8]: ArubaMC_GroupXX
Enter Switch Role (conductor|standalone|md) [md]: standalone
Enter Controller VLAN ID [1]: 98
Enter Controller VLAN port [GE 0/0/0]: [Enter]
Enter Controller VLAN port mode (access|trunk) [access]: trunk
Enter Native VLAN ID [1]: 98
Do you wish to configure IPV4 address on vlan (yes|no) [yes]: yes
Enter VLAN interface IP address [172.16.0.254]: 172.16.98.1-7 //เรียงตาม group
Enter VLAN interface subnet mask [255.255.255.0]: 255.255.255.0
Enter IP Default gateway [none]: 172.16.98.254
Enter DNS IP address [none]: 8.8.8.8
Do you wish to configure IPV6 address on vlan (yes|no) [yes]: no
Enter Country code (ISO-3166), <ctrl-I> for supported list: TH
You have chosen Country code TH for Thailand (yes|no)?: yes
Enter the controller's IANA Time zone [America/Los Angeles]: Asia/Bangkok
Enter Time in UTC [07:01:44]: 14:04:00 //Current Time HH:MM:SS
Enter Date (MM/DD/YYYY) [6/14/2025]: [Enter] // Check Date Current
Enter Password for admin login (up to 32 chars): P@ssw0rd
Re-type Password for admin login: P@ssw0rd
```

# Add License

Enter Web Browser URL >> [https://172.16.98.\(1-7\):4343](https://172.16.98.(1-7):4343)



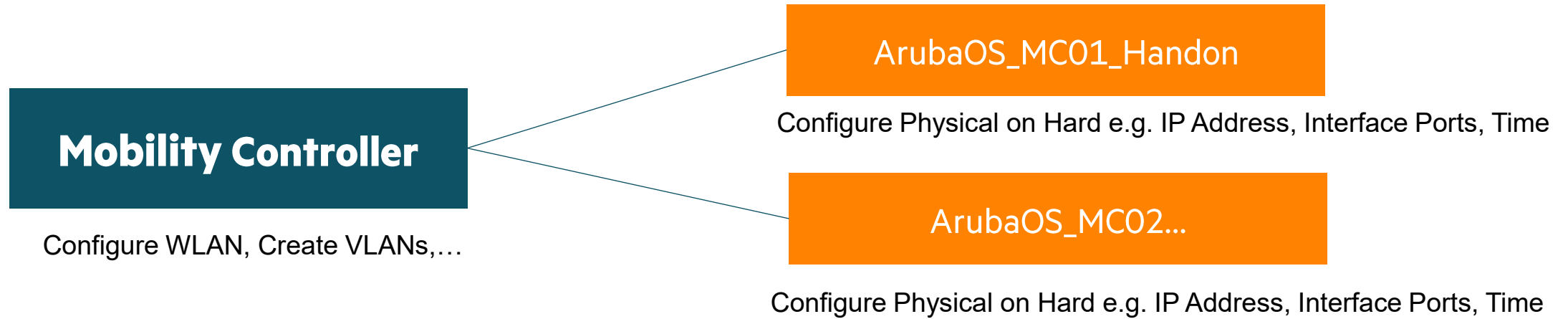
The screenshot shows a web browser window with the title "Login". The address bar displays "https://192.168.98.253:4343" with a "Not secure" warning. The main content area features the Aruba logo, the text "a Hewlett Packard Enterprise company", and the identifier "ArubaMC\_Group99". Below this are input fields for "Username" and "Password", a "Log in" button, and a copyright notice: "Copyright © 2024 Hewlett Packard Enterprise Development L.P.".

Username: **admin**

Password: **P@ssw0rd**



# Hierarchy



# Add License

**aruba** MOBILITY CONTROLLER ArubaMC\_Group99

ACCESS POINTS 0 0 CLIENTS 0 0 0 ALERTS 0

admin

**1** Mobility Controller

Dashboard

**2** Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

**3** License

Redundancy

Connect to external license server: ☐

**Usage**

| Usage Summary                | Usage by This Controller            |                                    | Usage by Other Controller                   |                              |                                     |                                     |
|------------------------------|-------------------------------------|------------------------------------|---------------------------------------------|------------------------------|-------------------------------------|-------------------------------------|
|                              | AP<br>Access Points                 | PEF<br>Policy Enforcement Firewall | RF Protect<br>Wireless Intrusion Protection | ACR<br>Advanced Cryptography | WebCC<br>Web Content Classification | VIA<br>Virtual Intranet Access      |
| Feature Enabled              | <input checked="" type="checkbox"/> | <input type="checkbox"/>           | <input type="checkbox"/>                    | <input type="checkbox"/>     | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Scope                        | Per-AP                              | Per-AP                             | Per-AP                                      | Per-Session                  | Per-AP                              | Per-Session                         |
| Licenses Installed           | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Expired Licenses             | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Active Licenses              | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Licenses Used                | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Licenses Remaining Available | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |

**4** > Inventory

# Set Time

**1**

Mobility Controller

**2**

ArubaMC\_Group99

Admin AirWave CPSec Certificates SNMP Logging Profiles

Dashboard

**Configuration**

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

**Basic Info**

Password for user admin:

Retype password:

> Domain Name System

# CPSec with auto cert provisioning

The screenshot displays the Aruba Mobility Controller web interface. The interface is divided into a left sidebar and a main content area. The sidebar contains a menu with categories: **Mobility Controller** (1), **Configuration** (2), and **System** (3). Under **Configuration**, there are links for WLANs, Roles & Policies, Access Points, AP Groups, Authentication, Services, and Interfaces. Under **System**, there are links for Tasks, License, and Redundancy. The main content area has a top navigation bar with tabs: General, Admin, AirWave, **CPsec** (4), Certificates, SNMP, Logging, and Profiles. A **Pending Changes** button with a refresh icon (6) is located in the top right corner. The **CPsec** tab is active, showing the **Control Plane Security** section. This section includes three toggle switches: **Enable CPsec:** (5), **Enable auto cert provisioning:** (5), and **Only accept APs from specified ranges:**. A link for **Cluster for Allowlist Propagation** is also visible.



# D. Configure VLANs

---

- Create VLANs
- Configure VLAN Interface
- Assign VLANs on port



# Create VLANs

The screenshot displays the Aruba Mobility Controller web interface. At the top, the header includes the Aruba logo, the system name 'ArubaMC\_Group99', and status indicators for Access Points (0), Clients (0), and Alerts (0). The user is logged in as 'admin'. The left sidebar shows a menu with 'Configuration' and 'Interfaces' highlighted. The main content area shows the 'VLANs' tab selected, with a table containing one entry (ID 1) and a '+ 5' button to add more.

**MOBILITY CONTROLLER**  
ArubaMC\_Group99

**ACCESS POINTS** 0 0  
**CLIENTS** 0 0 0  
**ALERTS** 0

**Mobility Controller**

Dashboard  
**Configuration**  
WLANs  
Roles & Policies  
Access Points  
AP Groups  
Authentication  
Services  
**Interfaces**  
System  
Tasks  
License  
Redundancy

Ports **VLANs** IP Routes GRE Tunnels Pool Management OSPF Multicast

**VLANs**

| NAME | ID(S) |
|------|-------|
| --   | 1     |

+ 5

# Configure VLAN Interface

Mobility Controller > **ArubaMC\_Group99**

Dashboard

**Configuration**

- WLANs
- Roles & Policies
- Access Points
- AP Groups
- Authentication
- Services
- Interfaces**
- System
- Tasks
- Redundancy
- IoT

Diagnostics

Maintenance

Ports **VLANs** Routes IPv6 Neighbors GRE Tunnels Pool Management

**VLANs**

| NAME   | ID(S)    |  |
|--------|----------|--|
| Client | 10,20,30 |  |
| --     | 1,98     |  |

**VLANs > Client**

| ID | IPV4 AD... | IPV6 AD... | ENABLE ... | PORT M... | ADMIN ... | OPERATI... | PD CLIE... | DHCP SE... |
|----|------------|------------|------------|-----------|-----------|------------|------------|------------|
| 10 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |
| 20 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |
| 30 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |

# Assign VLANs on port

Mobility Controller > ArubaMC\_Group99

Dashboard

**Configuration**

- WLANs
- Roles & Policies
- Access Points
- AP Groups
- Authentication
- Services
- Interfaces**
- System
- Tasks
- Redundancy
- IoT

Diagnostics

Maintenance

**Ports**

Port Channel

| NAME | MEMBERS | PROTOCOL | TRUSTED | POLICY | MODE | NATIVE VLAN | TRUNK VLANS |
|------|---------|----------|---------|--------|------|-------------|-------------|
| +    |         |          |         |        |      |             |             |

**Ports**

| PORT     | ADMI...  | TRUST... | POLICY   | MODE   | NATIV... | ACCES... | TRUN... | SPAN... | MONI... | DESC... |
|----------|----------|----------|----------|--------|----------|----------|---------|---------|---------|---------|
| GE-0/... | Enabl... | ✓        | Not-d... | trunk  | 98       | 1        | 1-4094  | ✓       | --      | --      |
| GE-0/... | Enabled  | ✓        | Not-d... | access | 1        | 1        | 1-4094  | ✓       | --      | --      |
| GE-0/... | Enabled  | ✓        | Not-d... | access | 1        | 1        | 1-4094  | ✓       | --      | --      |
| GE-0/... | Enabled  | ✓        | Not-d... | access | 1        | 1        | 1-4094  | ✓       | --      | --      |

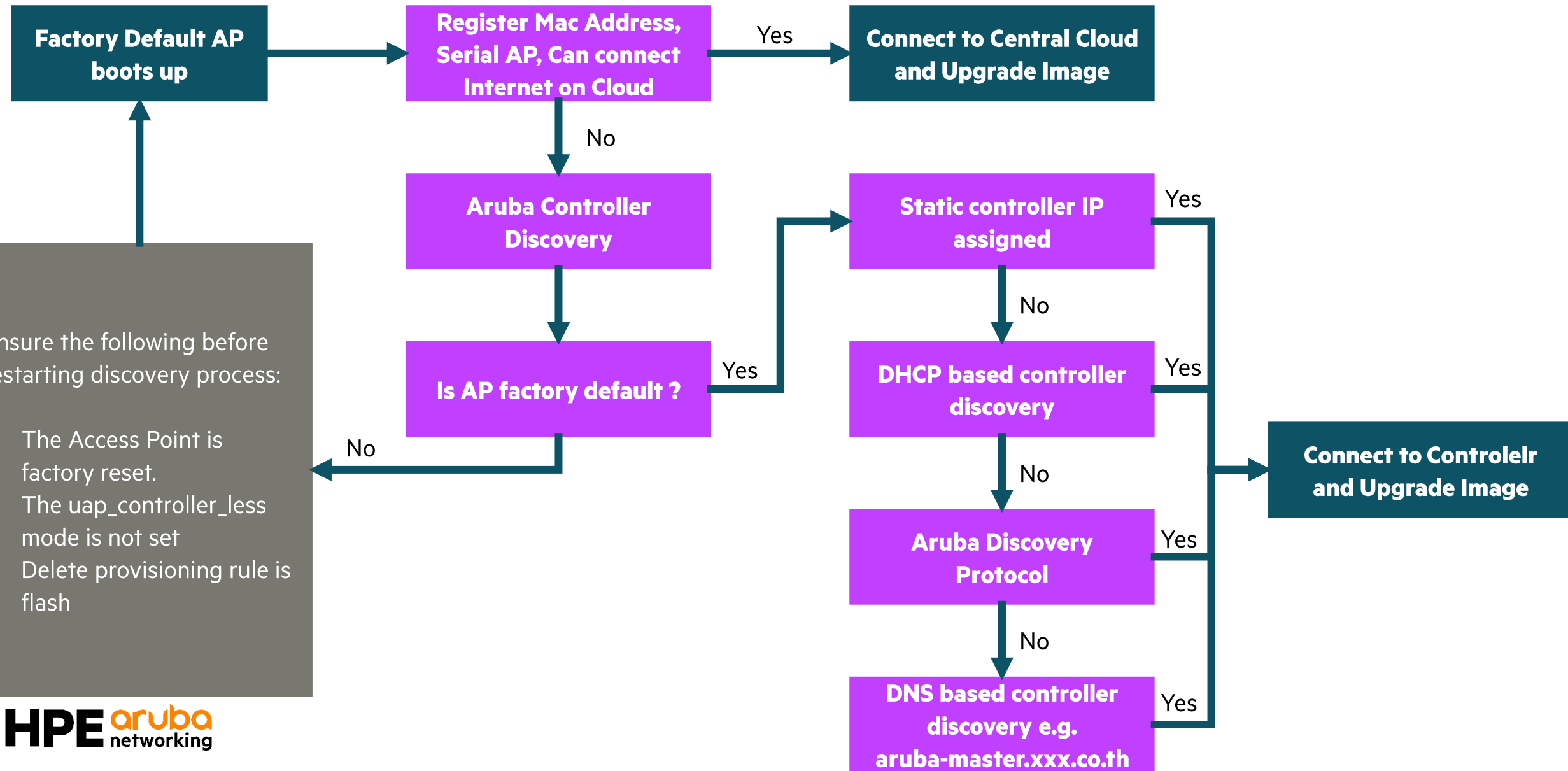


# **E.Initial Access Point(AP)**

- **Understand Flow Unified**
- **LED Status**
- **Reset Setup**
- **AP convert to Controller**

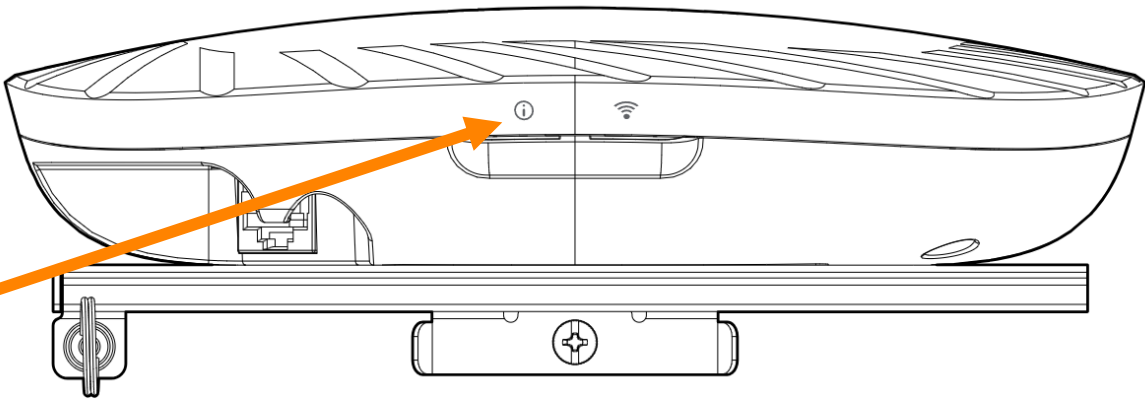


# Understand Flow Unified AP





# LED Status

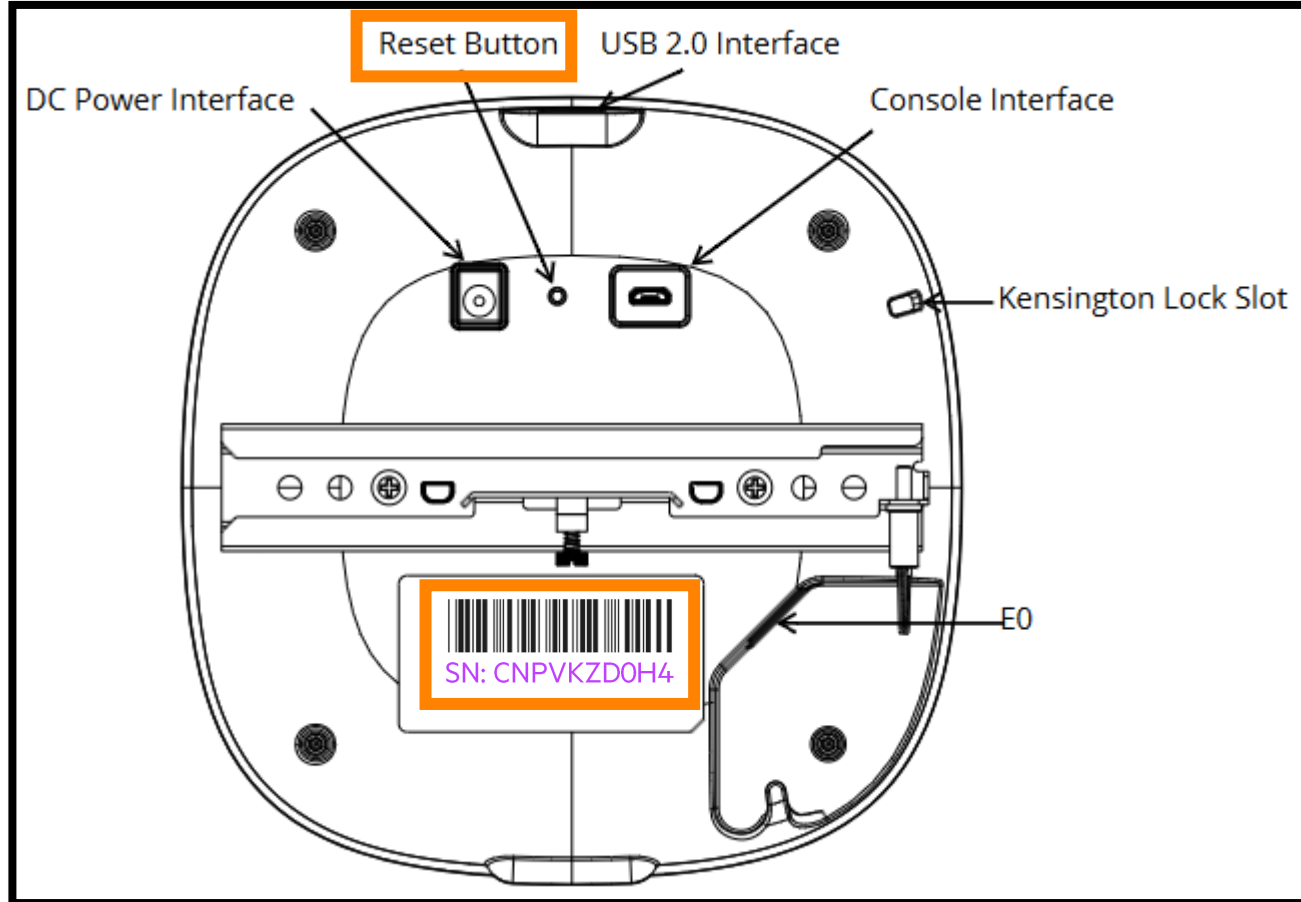


System Status LED ⓘ

| Color/State                    | Meaning                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Off                            | Device Powered off                                                                                              |
| Green - solid                  | Device ready, fully functional, no network restrictions                                                         |
| Green - blinking <sup>1</sup>  | Device booting, not ready                                                                                       |
| Green - flash on <sup>2</sup>  | Device ready, fully functional, uplink negotiated in sub-optimal speed (<1Gbps)                                 |
| Green - flash off <sup>3</sup> | Device in deep-sleep mode                                                                                       |
| Amber - solid                  | Device ready, restricted power mode (IPM restrictions applied), no network restrictions                         |
| Amber - flashing off           | Device ready, restricted power mode (IPM restrictions applied), uplink negotiated in sub-optimal speed (<1Gbps) |
| Red                            | System error conditions – Immediate action required                                                             |

- 1. Blinking: one second on, one second off, 2 seconds cycle.
- 2. Flashing off: mostly on, fraction of a second off, 2 seconds cycle.
- 3. Flashing on: mostly off, fraction of a second on, 2 seconds cycle.

# Reset Setup



## Opt-1 Reset the AP during normal operation

press and hold down the reset button using a small, narrow object such as a paper clip for more than 10 seconds during normal operation.

## Opt-2 Reset the AP while powering up

1. Press and hold down the reset button using a small, narrow object such as a paper clip while the access point is not powered on (either via DC power or PoE).
2. Connect the power supply (DC or PoE) to the access point while the reset button is being held down.
3. Release the reset button on the access point after 15 seconds.

## Next Step

**Access Point broadcast SSID : SetMeUp-XX:XX:XX**

 SetMeUp-C2:69:D4

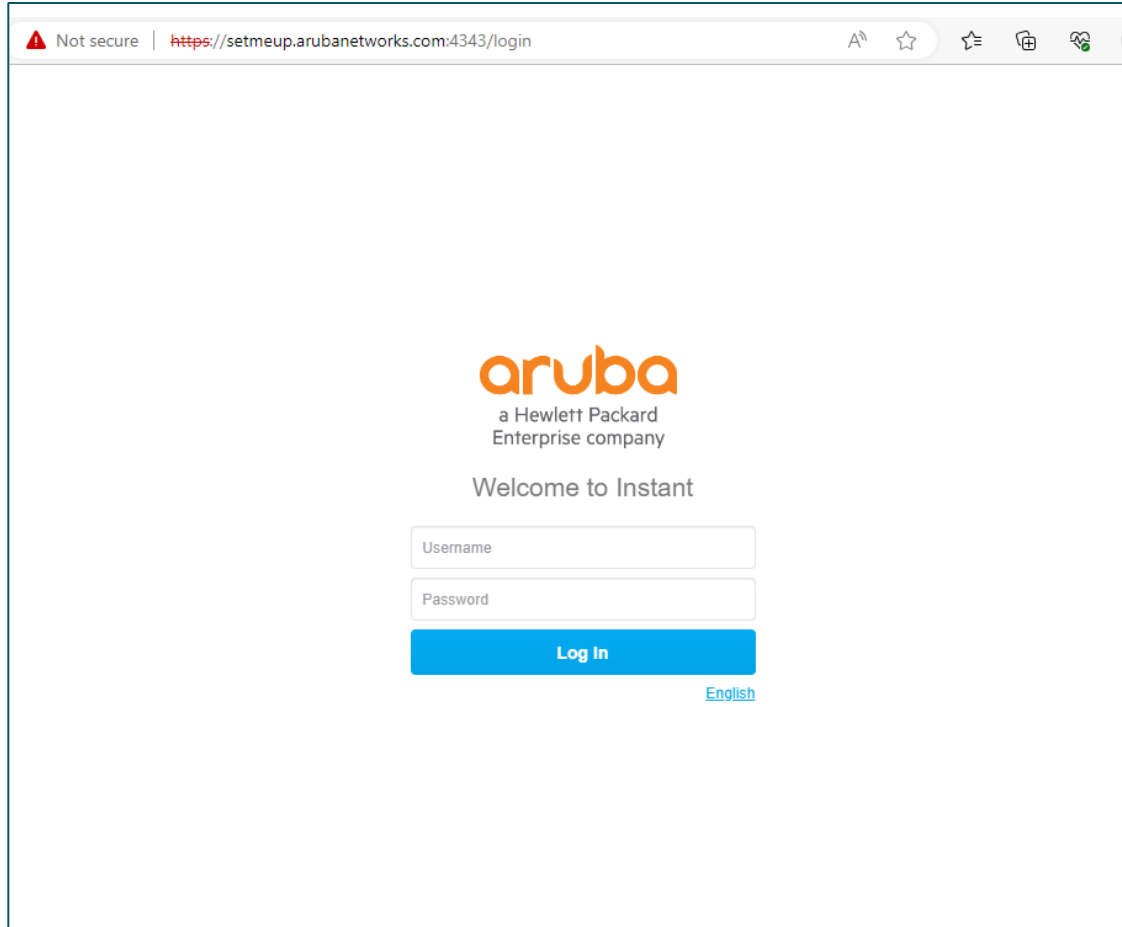
## Default User/Pass

Username : **admin** Password : **CNPVKZD0H4** //S/N (Uppercase)

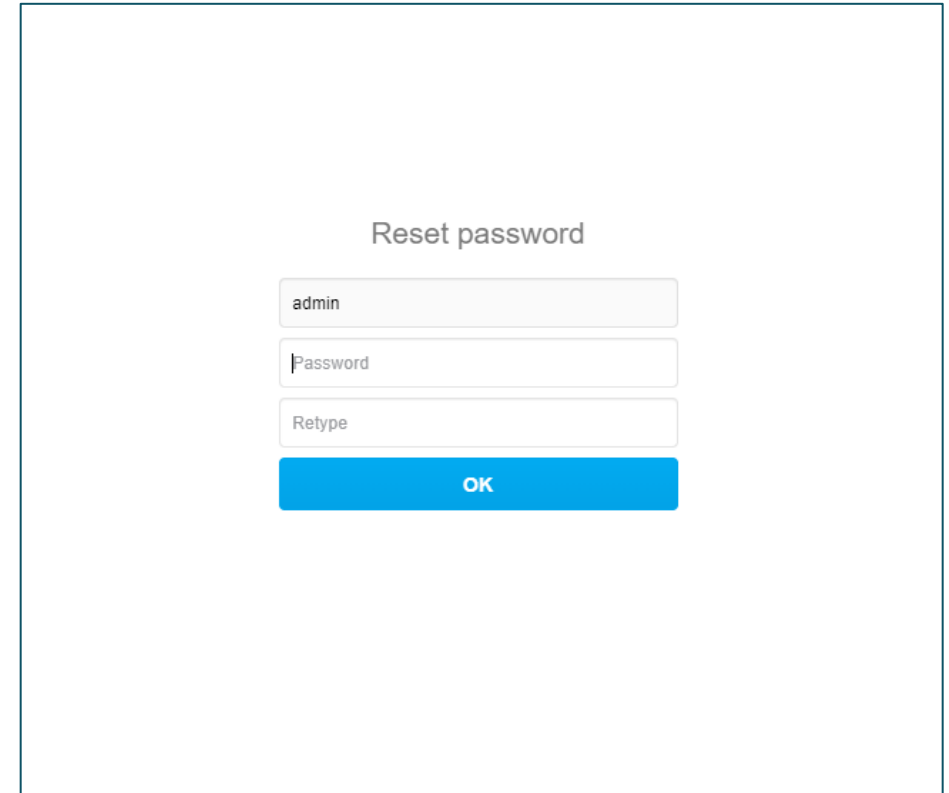
# Login GUI and Changes password

## Default User/Pass

Username : **admin** Password : **CNPVKZD0H4** //S/N (Uppercase)



A screenshot of a web browser showing the Aruba login page. The browser's address bar displays "https://setmeup.arubanetworks.com:4343/login". The page features the Aruba logo (a Hewlett Packard Enterprise company) and the text "Welcome to Instant". Below this, there are two input fields: "Username" and "Password", followed by a blue "Log In" button. A small "English" link is visible at the bottom right of the login area.

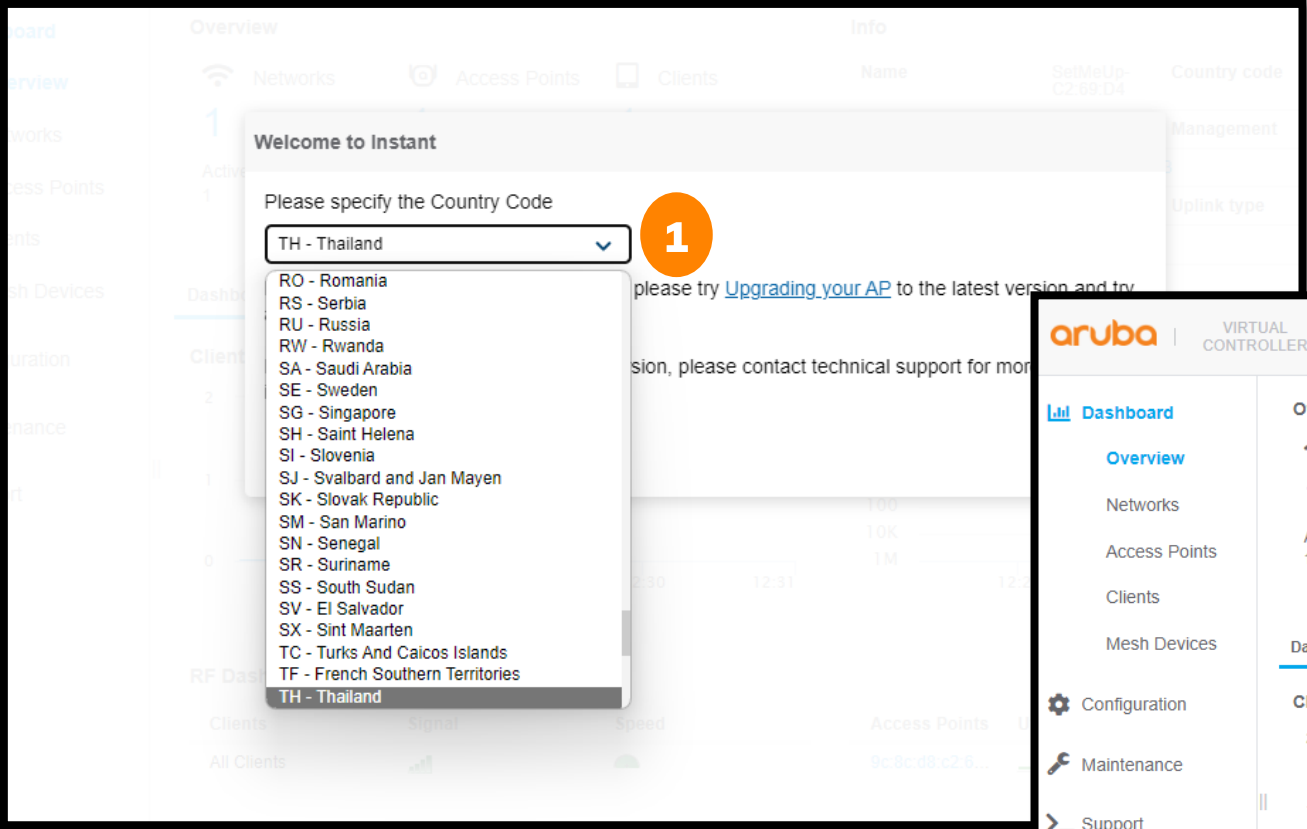


A screenshot of the "Reset password" page. It contains three input fields: the first is pre-filled with "admin", the second is labeled "Password", and the third is labeled "Retype". Below these fields is a blue "OK" button.

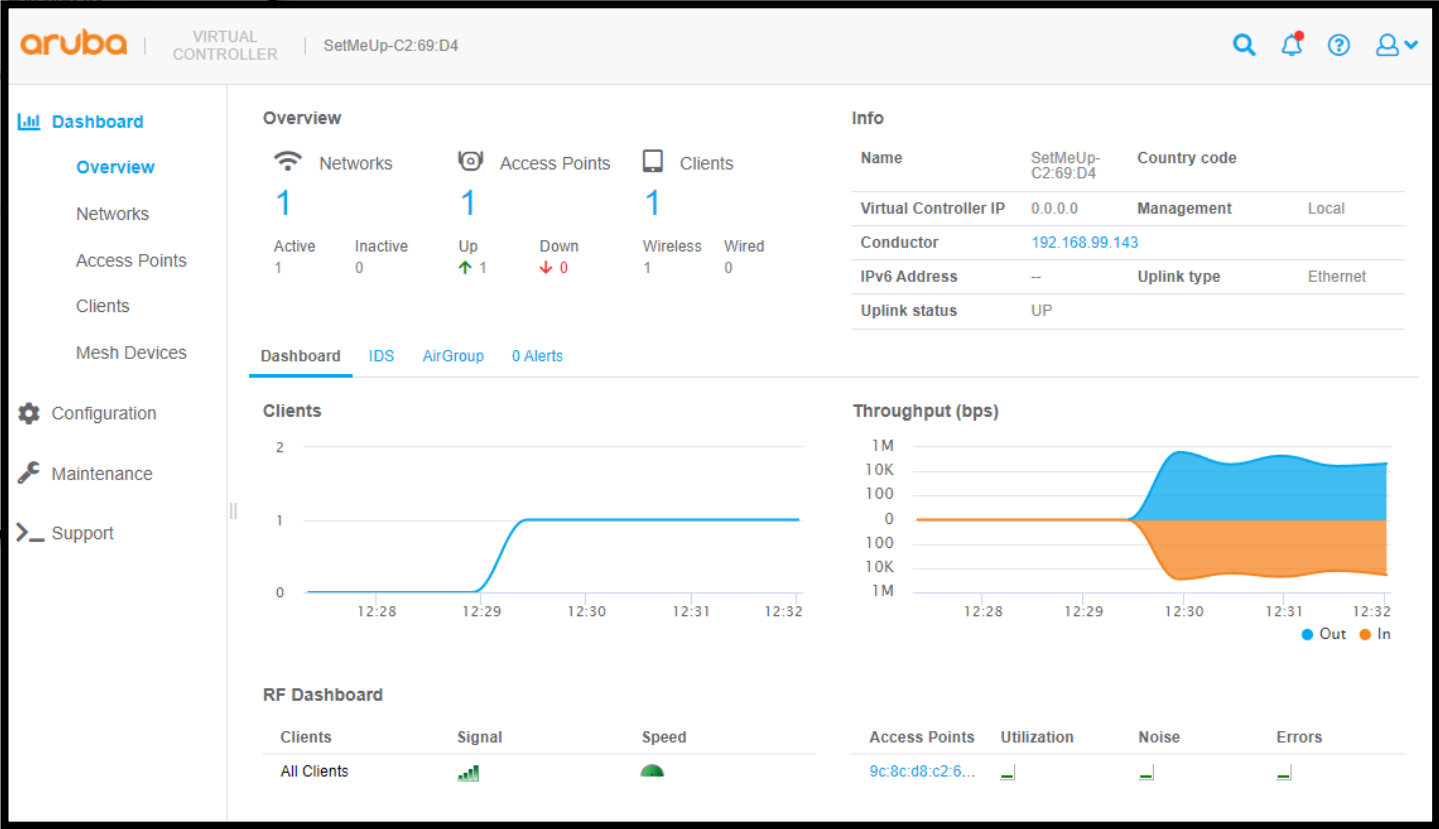
Enter new **Password: P@ssw0rd** and **Retype password: P@ssw0rd**

# Select Country Code

Select : **TH-Thailand**



**Dashboards-> Overview**



# Convert Access Point Campus mode

aruba | VIRTUAL CONTROLLER | SetMeUp-C2:69:D4

Dashboard

- Overview
- Networks
- Access Points
- Clients
- Mesh Devices

Configuration

**Maintenance** 1

- About
- Firmware
- Configuration
- Certificates
- Reboot

**Convert**

Convert one or more Access Points to

Hostname or IP Address of Mobility Controller

After conversion, all Access Points will be managed by the Controller specified above.

**Convert**

Campus APs managed by a Mobility Controller 1

- Remote APs managed by a Mobility Controller
- Campus APs managed by a Mobility Controller
- Standalone AP
- Single AP

aruba | VIRTUAL CONTROLLER | SetMeUp-C2:69:D4

Dashboard

- Overview
- Networks
- Access Points
- Clients
- Mesh Devices

Configuration

**Maintenance** 1

- About

**Convert**

Convert one or more Access Points to

Hostname or IP Address of Mobility Controller

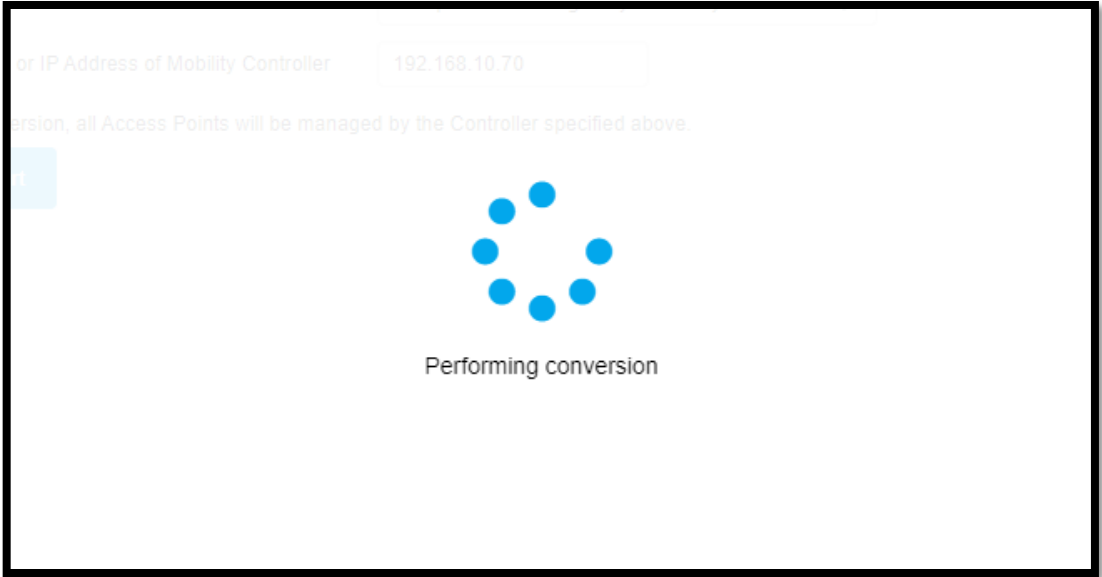
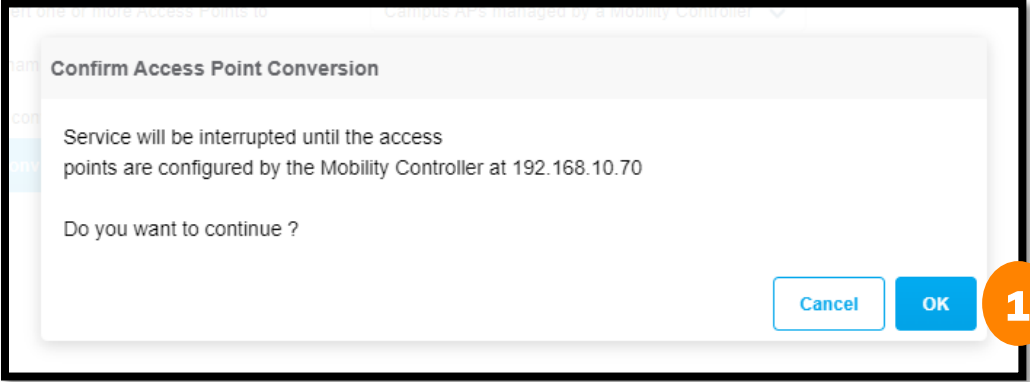
172.16.98.1-7

After conversion, all Access Points will be managed by the Controller specified above.

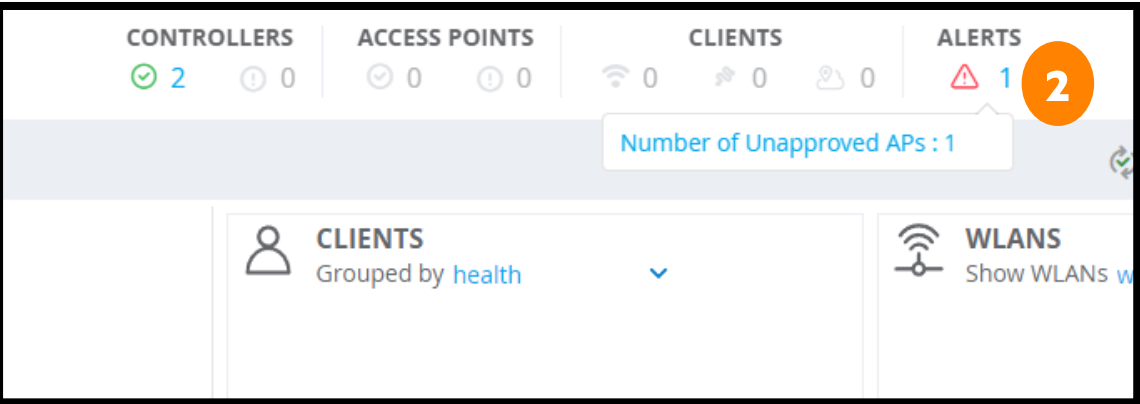
**Convert** 1



# Convert Access Point Campus mode (Cont.)



Approve device on Mobility Controller(MC)  
**In-Case: enable auto cert provisioning not yet.**



\_\_\_\_\_

Top bar show one **ALERTS** “Number of Unapproved APs : 1”

The dashboard provides a high-level overview of the network's health. It features four main status sections at the top: Controllers (2 approved, 0 unapproved), Access Points (0 approved, 0 unapproved), Clients (0 connected, 0 disconnected), and Alerts (1 active). A prominent orange callout box highlights the 'Number of Unapproved APs : 1'. Below these sections, there are expandable cards for 'CLIENTS' (grouped by health) and 'WLANS' (showing WLANs with 100% up time).

| CONTROLLERS | ACCESS POINTS | CLIENTS | ALERTS |
|-------------|---------------|---------|--------|
| 2           | 0  0          | 0  0    | 1      |

Number of Unapproved APs : 1

**CLIENTS**  
Grouped by health

**WLANS**  
Show WLANs with 100% up time

Campus APs
Remote APs
Mesh APs
**Allowlist**
Provisioning Rules

Campus AP Allowlist

Remote AP Allowlist

| <input checked="" type="checkbox"/> | MAC ADDRESS       | NAME | AP GROUP | DESCRIPTION | STATUS   | REVOKE TEXT | APPROVED | UPDATED          |
|-------------------------------------|-------------------|------|----------|-------------|----------|-------------|----------|------------------|
| <input checked="" type="checkbox"/> | 9c:8c:d8:c2:69:d4 | --   | --       | --          | Accepted | --          | No       | Mon Jul 10 12:40 |

+

Delete
Revoke
Approve

50
1



# F. Create AP Group

---

- Provision AP Change to Group



# Create Group AP

aruba

MOBILITY  
CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS

✓ 1 ⓘ 0

CLIENTS

📶 0 📶 0 📶 0

ALERTS

⚠ 0

?

admin ▾

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

AP Groups 2

| NAME          | APs |
|---------------|-----|
| default       | 1   |
| NoAuthApGroup | --  |

+

New AP Group

Name:

Cancel

Submit

# Change AP to New Group

Mobility Controller > 1

Dashboard

Configuration 2

WLANs

Roles & Policies

Access Points 3

AP Groups

Authentication

Services

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Tasks

License

Redundancy

Campus APs Remote APs Mesh APs Allowlist Provisioning Rules

Campus APs 1

| <input checked="" type="checkbox"/> | AP NAME       | AP GROUP | IPV4 ADD...  | IPV6 ADD... | SWITCH IP    | MAC ADD...    | SERIAL #   | T |
|-------------------------------------|---------------|----------|--------------|-------------|--------------|---------------|------------|---|
| <input checked="" type="checkbox"/> | 7c:57:3c:c... | default  | 192.168.9... | --          | 192.168.9... | 7c:57:3c:c... | CNPBK51... | 3 |

5 Provision

Flags:  
U = Unprovisioned, N = Duplicate name, G = No such group, L = Unice...  
Mismatch, X = Maintenance Mode, P = PPPoE AP, B = Built-in AP, s = LA...  
RAP, c = CERT-based RAP, 1 = 802.1x authenticated AP use EAP-PEAP, 1...  
u = Custom-Cert RAP, S = Standby-mode AP, J = USB cert at AP, f = No S...  
Mesh Recovery, z = Datazone AP, e = Custom EST cert, p = In deep-sleep...  
ShutDown, F = AP failed 802.1x authentication

7c:57:3c:c3:e0:e4

MAC address: 7c:57:3c:c3:e0:e4

LLDP neighbor chassis ID / port ID: 6200 / 1/1/3

Name: 7c:57:3c:c3:e0:e4

AP group: Group-99 6 Select New Group

Controller discovery: ☒ Use AP discovery protocol (ADP) ☐ Static

Controller discovery preference: ☒ IPv4 ☐ IPv6

IP: ☒ DHCP ☐ Static

Deployment: ☒ Campus ☐ Remote ☐ Mesh ☐ Remote mesh portal

Wi-Fi uplink: ☐ Wi-Fi uplink

advanced options

Cancel Submit 6

Access Points will be Rebooted

CAUTION: Applying this configuration change will interrupt service while the affected Access Points are rebooted.

Do you want to continue ?

Cancel Continue & Reboot 7



# A. Configure WLAN PSK+MAC

- Add local user

# Configure WLAN PSK

aruba

MOBILITY  
CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

👤 0

📍 0

ALERTS

⚠ 0

?

admin ▾

Mobility Controller

1

Dashboard

Configuration

2

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

WLANs 0

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |  |
|-------------|----------|----------------|-------------|--|
| + 3         |          |                |             |  |



\_\_\_\_\_



# Configure WLAN PSK

aruba

MOBILITY CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS

1

0

CLIENTS

1

0

0

ALERTS

0

admin

Mobility Controller

Search

Dashboard

Overview

Infrastructure

Traffic Analysis

Security

Services

Configuration

Diagnostics

Maintenance

Client

WLAN

265 kB

Radios

Wireless Clients 1

| NAME         | IP ADDRESS   | HEALTH | BAND  | ROLE          | SNR   | USAGE  | WLAN    | CONNECTED...      |
|--------------|--------------|--------|-------|---------------|-------|--------|---------|-------------------|
| 192.168.20.1 | 192.168.20.1 | Good   | 5 GHz | authenticated | 54 dB | 104 kB | PSK-G99 | 7c:57:3c:c3:e0:e4 |

DETAILS

Name  
192.168.20.1

IP address  
192.168.20.1

MAC address  
a4:d9:31:b8:57:75

Health score  
92%

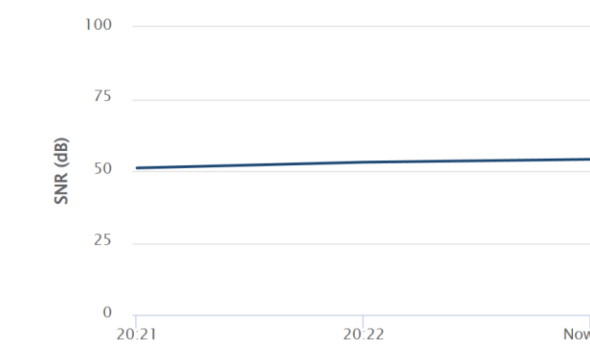
Speed  
842 Mbps

Max speed  
866 Mbps

Frames in the last minute  
121

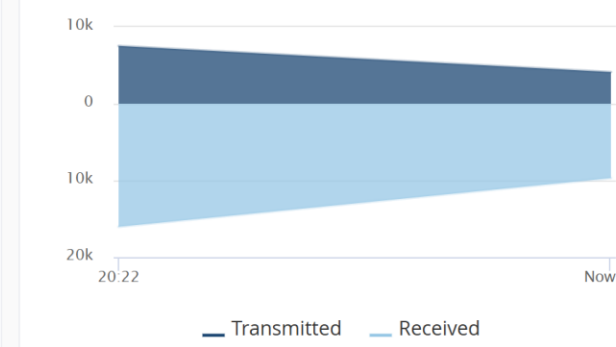
SIGNAL

Show information about signal quality



USAGE

Show information about throughput



2

Copy mac address

# Add mac address to local user

aruba

MOBILITY CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS  
1 0

CLIENTS  
1 0 0

ALERTS  
0

admin

Mobility Controller

Dashboard

Configuration1

WLANs

Roles & Policies

Access Points

AP Groups

Authentication2

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Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Auth Servers3

Profiles

L2 Authentication

L3 Authentication

User Rules

Advanced

Server Groups 2

| NAME     | SERVICES | FAIL THROUGH | LOAD BALANCE | SERVICES RULES |   |
|----------|----------|--------------|--------------|----------------|---|
| default  | 1        | --           | --           | 1              |   |
| internal | 1        | --           | --           | 1              | 4 |

+

Server Group > internal

ServersOptionsServer Rules

Drag rows to re-order

| NAME     | TYPE | IP ADDRESS | TRIM FQDN | MATCH RULES |   |
|----------|------|------------|-----------|-------------|---|
| Internal | --   | --         | --        | 0           | 5 |

+

Server Group > internal > Internal

UsersOptionsImportExportServer Group Trim FQDNServer Group Match RulesGuest user page

| USER NAME | ROLE | EMAIL | EXPIRATION | STATIC IP FOR RAPS | ENABLED |  |
|-----------|------|-------|------------|--------------------|---------|--|
|-----------|------|-------|------------|--------------------|---------|--|

+

6



# Add mac address to local user

1 Past mac address to username, password, retype password

Internal Server > Add New User

|                           |                                            |                                         |
|---------------------------|--------------------------------------------|-----------------------------------------|
| User name:                | <input type="text" value="a4d931b85775"/>  | <input type="button" value="Generate"/> |
| Password:                 | <input type="password" value="....."/>     | <input type="button" value="Generate"/> |
| Retype password:          | <input type="password" value="....."/>     |                                         |
| Role:                     | <input type="text" value="authenticated"/> | 2                                       |
| E-mail:                   | <input type="text"/>                       |                                         |
| Static inner IP for RAPs: | <input type="text"/>                       |                                         |
| Enabled:                  | <input checked="" type="checkbox"/>        |                                         |
| Expiration:               | <input type="text" value="None"/>          |                                         |

Cancel Submit 3

# Configure WLAN Add MAC Authentication

1. The **Mobility Controller** menu item is highlighted in the top navigation bar.

2. The **WLANs** menu item is highlighted in the left sidebar under the **Configuration** section.

3. The **WLANs** table is shown with the following data:

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| PSK-G99     | default  | WPA2-Personal  | --          |

A blue plus icon is visible below the table to add a new WLAN.

4. The **Security** tab is selected for the **PSK-G99** WLAN. The configuration shows:

- Key management:** WPA3-Personal (dropdown)
- ☒ **Enable backward compatibility**
- Passphrase:** [Redacted]
- Retype:** [Redacted]
- MAC authentication:** Enabled (dropdown) - This option is highlighted with an orange box.
- Denylisting:** [Off]

A security slider on the left is positioned at **Personal**, between **Enterprise** and **Open**. The slider scale ranges from **More Secure** at the top to **Less Secure** at the bottom.

**4 Enable MAC authentications**

Buttons: **Cancel** and **Submit**

# Configure WLAN Add MAC Authentication

PSK-G99   General   VLANs   Security   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Cancel** **Submit**

**Pending Changes**

**WLANs 1**

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| PSK-G99     | default  | WPA2-Personal  | --          |

**General**   **VLANs**   **Security**   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Pending Changes**

☒ Pending Changes for 1 Group

☒ [+](#) Mobility Controller

**Close** **Discard changes** **Deploy changes**

# Configure WLAN Add MAC Authentication

PSK-G99   General   VLANs   Security   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Cancel** **Submit**

**Pending Changes**

**WLANs 1**

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| PSK-G99     | default  | WPA2-Personal  | --          |

**General**   **VLANs**   **Security**   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Pending Changes**

☒ Pending Changes for 1 Group

☒ [+](#) Mobility Controller

**Close** **Discard changes** **Deploy changes**



# B. Configure WLAN 802.1X

---



# Configure WLAN 802.1X

aruba

MOBILITY  
CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

👤 0

📍 0

ALERTS

⚠️ 0

?

admin ▾

☰

Mobility Controller

1

Dashboard

Configuration

WLANs

2

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

WLANs 0

| NAME (SSID)    | AP GROUP | KEY MANAGEMENT | INFORMATION |  |
|----------------|----------|----------------|-------------|--|
| <div>+ 3</div> |          |                |             |  |

# Configure WLAN 802.1X

**New WLAN**

General VLANs Security Access

Name (SSID): 802.1X-G99 **1**

Primary usage: ☒ Employee ☐ Guest

Select AP Groups **2**

Broadcast on: ☐ default ☒ Group-99

Forwarding mode: Tunnel

Cancel Back Next **3**

**New WLAN**

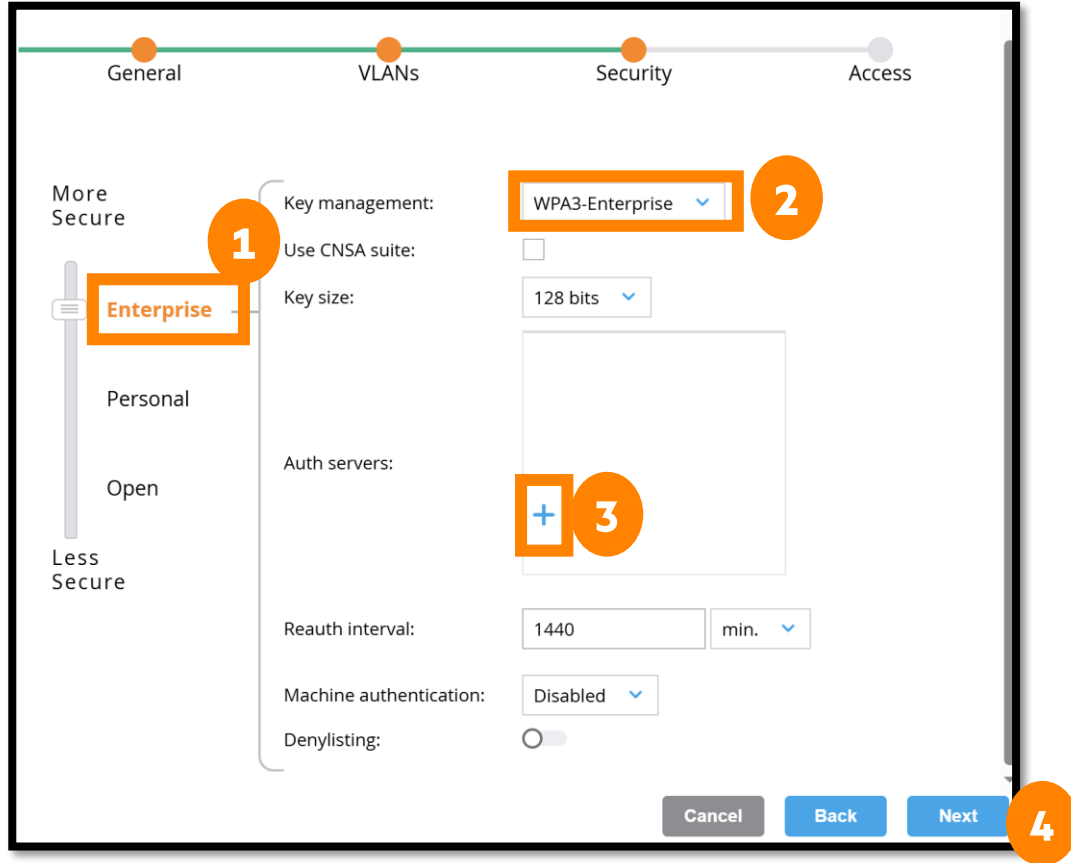
General VLANs Security Access

VLAN: 10 **4** Select vlan 10

[Show VLAN details](#)

Cancel Back Next **5**

# Configure WLAN 802.1X



General VLANs Security Access

More Secure

Enterprise (1)

Personal

Open

Less Secure

Key management: WPA3-Enterprise (2)

Use CNSA suite: ☐

Key size: 128 bits

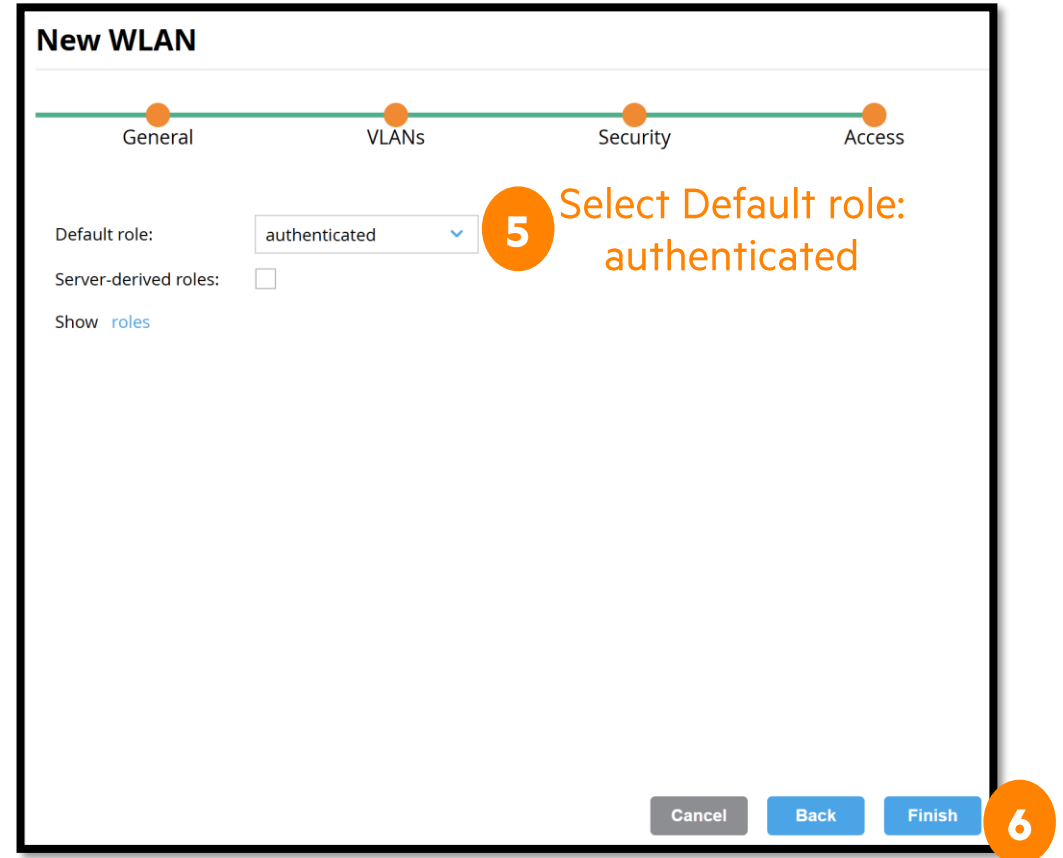
Auth servers: + (3)

Reauth interval: 1440 min.

Machine authentication: Disabled

Denylisting: ☐

Cancel Back Next (4)



New WLAN

General VLANs Security Access

Default role: authenticated (5)

Server-derived roles: ☐

Show roles

Cancel Back Finish (6)

# Configure add local user for 802.1X

**Mobility Controller**

Dashboard  
**Configuration** (1)  
WLANs  
Roles & Policies  
Access Points  
AP Groups  
**Authentication** (2)  
Services  
Interfaces  
System  
Tasks  
License  
Redundancy  
Diagnostics  
Maintenance

**Auth Servers** (3) AAA Profiles L2 Authentication L3 Authentication

**Server Groups** 3

| NAME              | SERVICES | FAIL THROUGH | LOAD BALANCE | SERVER RULES |
|-------------------|----------|--------------|--------------|--------------|
| default           | 1        | --           | --           | 1            |
| Internal          | 1        | --           | --           | 1            |
| 802.1X-G99_dot... | 1        | --           | --           | 0            |

**All Servers** 1

| NAME     | TYPE | IP ADDRESS / HOST | SERVER GROUP             |
|----------|------|-------------------|--------------------------|
| Internal | --   | --                | default internal 802.... |

**Auth Servers** AAA Profiles L2 Authentication L3 Authentication

**Server > Internal** **Users** Options Import Export [Guest user page](#)

| USER NAME     | ROLE          | EMAIL | EXPIRATION | STATIC IP FO... | ENABLED |
|---------------|---------------|-------|------------|-----------------|---------|
| a4d931b857... | authenticated | --    | --         | 0.0.0.0         | Yes     |

(5) +

50 < 1 >

**Internal Server > Add New User** (6)

User name: group99 **Generate**

Password: ..... **Generate**

Retype password: .....

Role: authenticated (7)

E-mail:

Static inner IP for RAPs:

Enabled: ☒

Expiration: None

Create User  
User: group0X  
Pass: P@sswOrd

Cancel Submit (8)



# C. Configure WLAN Guest

- Add local user



# Configure WLAN Guest

The screenshot shows the Aruba Mobility Controller web interface. At the top, the Aruba logo is on the left, and the title 'MOBILITY CONTROLLER ArubaMC\_Group99' is in the center. To the right of the title are three status sections: 'ACCESS POINTS' with a green checkmark and '1', 'CLIENTS' with a Wi-Fi icon and '0', and 'ALERTS' with a warning triangle and '0'. Further right is a help icon and a user dropdown menu showing 'admin'. Below the title bar is a navigation bar with a hamburger menu icon and the text 'Mobility Controller'. A sidebar on the left contains a 'Dashboard' link and a 'Configuration' link, which is highlighted with an orange box and a '2'. Under 'Configuration' is a 'WLANs' link. The main content area shows a table titled 'WLANs 0'. The table has four columns: 'NAME (SSID)', 'AP GROUP', 'KEY MANAGEMENT', and 'INFORMATION'. Below the table is a large '+3' button, which is highlighted with an orange circle and a '3'.

aruba

MOBILITY CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS  
✓ 1 0

CLIENTS  
0 0 0

ALERTS  
0

admin

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

WLANs 0

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| + 3         |          |                |             |

# Configure WLAN Guest

## New WLAN

General VLANs Security Access

Name (SSID):  1

Primary usage: ☐ Employee ☒ Guest 2

Select AP Groups

Broadcast on: ☐ default ☒ Group-99 3

Forwarding mode:

Cancel Back Next 4

## New WLAN

General VLANs Security Access

VLAN:  5

Select vlan 30

[Show VLAN details](#)

Cancel Back Next 6



# **D. Configure Role & Policy**

- **Network aliases**



# Network Aliases

Configure on node “Mobility Controller”-> Configuration-> Roles & Policies -> Aliases -> +

The screenshot displays the Aruba Mobility Conductor web interface. At the top, the header includes the Aruba logo, the system name "MOBILITY CONDUCTOR ArubaOS\_MM01\_Ha...", and status indicators for Controllers (2 online, 0 offline), Access Points (0 online, 0 offline), Clients (1 online, 0 offline), and Alerts (0). A user menu shows "admin" with a dropdown arrow.

The left sidebar contains a navigation menu with the following items: Dashboard, Configuration (highlighted), WLANs, Roles & Policies (highlighted), Access Points, AP Groups, Authentication, Services, Interfaces, Controllers, System, and Tasks.

The main content area is titled "Mobility Controller" and features a sub-navigation bar with "Roles", "Policies", "Applications", and "Aliases" (which is selected and underlined). Below this, a section titled "Network Aliases" contains a table with the following data:

| NAME        | ITEMS | DESCRIPTION | IP VERSION | INVERT |  |
|-------------|-------|-------------|------------|--------|--|
| mswitch     | 1     | --          | IPv4       | --     |  |
| controller  | 1     | --          | IPv4       | --     |  |
| vrrp_ip     | 1     | --          | IPv4       | --     |  |
| any         | 1     | --          | IPv4       | --     |  |
| user        | 1     | --          | IPv4       | --     |  |
| localip     | 1     | --          | IPv4       | --     |  |
| auth-google | 2     | --          | IPv4       | --     |  |

Below the table is a blue plus sign icon (+) to add a new alias. A vertical scrollbar is visible on the right side of the table.

# Network Aliases (cont.)

Enter Name "Local\_Network" +

**Destination Local\_Network**

IP version: IPv4

Name: Local\_Network

Description:

Invert: ☐

Items

| TYPE | IP ADDRESS | NETMASK/RANGE/... | DOMAIN NAME |
|------|------------|-------------------|-------------|
|------|------------|-------------------|-------------|

+

Rule type : **Network**  
IP Address : **192.168.0.0**  
Network mask : **255.255.0.0**  
**OK -> Submit**

**Add New Destination**

Rule type: Network

IP address: 192.168.0.0

Network mask: 255.255.0.0

Cancel

OK



# Policies

## Create Policies

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Roles

Policies

Applications

Aliases

| NAME           | RULES COU... | TYPE    | POLICY US...     | DESCRIPTION |
|----------------|--------------|---------|------------------|-------------|
| validuseret... | 1            | eth     | --               | --          |
| global-sacl    | 0            | session | guest, statef... | --          |
| sdn-acl        | 0            | session | --               | --          |
| sys-control    | 13           | session | sys-ap-role, ... | --          |
| sys-ap-acl     | 12           | session | sys-ap-role      | --          |
| switch-logo... | 1            | session | switch-logon     | --          |
| sys-switch-acl | 12           | session | sys-switch-r...  | --          |

+

New Policy

Policy type:

Session

Policy name:

Local\_Network-Polic

Description:

Cancel

Submit

# Policies (cont.)

Roles **Policies** Applications Aliases

**Policies**

| NAME                          | RULES C... | TYPE           | POLICY ...   | DESCRIP... |          |
|-------------------------------|------------|----------------|--------------|------------|----------|
| apprf-sys-ap-role-sacl        | 0          | session        | sys-ap-role  | --         |          |
| apprf-sys-switch-role-sacl    | 0          | session        | sys-switc... | --         |          |
| apprf-guest-99-guest-log...   | 0          | session        | Guest-99...  | --         |          |
| <b>Local_Network-Policies</b> | 0          | <b>session</b> | --           | --         | <b>1</b> |
| uplink-lb-cfg-racl            | 0          | routing        | --           | --         |          |
| uplink-lb-sys-racl            | 0          | routing        | --           | --         |          |
| master-boc-traffic            | 0          | routing        | --           | --         |          |

**Policy > Local\_Network-Policies Rules** ⓘ Drag rows to re-order

| IP VERSI... | SOURCE | DESTINA... | SERVICE/... | ACTION | DESCRIP... |  |
|-------------|--------|------------|-------------|--------|------------|--|
|-------------|--------|------------|-------------|--------|------------|--|

**2**

## New Rule for Local\_Network-Policies

Rule type: ☒ Access control **3** ☐ Application

Cancel

OK **4**

## Local\_Network-Policies > New forwarding Rule

IP version: IPv4 **5**  
Source: Any  
Destination: Alias  
Destination alias: local\_network  
Service/app: Service  
Services alias: svc-icmp  
Action: Deny

TOS:  
Time range: - None - Reset  
802.1p priority:  
Options:  
☐ Log ☐ Mirror ☐ Denylist ☐ Disable scanning  
Queue:  
Description:

Cancel

Submit **6**

Create Policies Block ping

IP Version : **IPV4**

Source : **Any**

Destination : **Alias**

Destination alias : **Local\_Network**

Service/app : **Service**

Service alias : **svc-icmp**

Action : **Deny**

**Submit**

# Role

## Create Role

Mobility Controller

Dashboard

Configuration

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Roles

Policies

Applications

Aliases

Roles 15

| NAME            | RULES    |
|-----------------|----------|
| logon           | 32 Rules |
| guest           | 11 Rules |
| ap-role         | 35 Rules |
| stateful-dot1x  | 0 Rules  |
| guest-logon     | 27 Rules |
| sys-ap-role     | 25 Rules |
| sys-switch-role | 25 Rules |

+



New Role

Name:

engineer

Cancel

Submit

# Role

## Assign policies in Role engineer

Mobility Controller >

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Roles

ies

Applications

Aliases

Roles 16

| NAME                 | RULES    |
|----------------------|----------|
| denyall              | 1 Rules  |
| default-via-role     | 3 Rules  |
| default-vpn-role     | 4 Rules  |
| authenticated        | 4 Rules  |
| voice                | 45 Rules |
| Guest-99-guest-logon | 26 Rules |
| engineer             | 0 Rules  |

+

engineer

Show Advanced View

Global Rules

| IP VERSI... | SOURCE | DESTINA... | SERVICE/... | ACTION | DESCRIP... |
|-------------|--------|------------|-------------|--------|------------|
|-------------|--------|------------|-------------|--------|------------|

+

engineer

Policies

Bandwidth

Captive Portal

Show Basic View

| NAME           | RULES COU... | TYPE    | POLICY US...    | DESCRIPTI... |
|----------------|--------------|---------|-----------------|--------------|
| global-sacl    | 0            | session | guest, state... | --           |
| apprf-engin... | 0            | session | engineer        | --           |
| engineer       | 0            | session | engineer        | --           |

+

# Role

Assign policies > local\_network(first), allowall(second)

**New Policy**

☒ Add an existing policy **1** ☐ Create a new policy

Policy type: Session

Policy name: Local\_Network-Policies **2**

Description:

Position:

**3** Cancel Submit

first  
→

| engineer       | Policies     | Bandwidth | Captive Portal  | Show Basic View |
|----------------|--------------|-----------|-----------------|-----------------|
| NAME           | RULES COU... | TYPE      | POLICY US...    | DESCRIPTI...    |
| global-sacl    | 0            | session   | guest, state... | --              |
| apprf-engin... | 0            | session   | engineer        | --              |
| engineer       | 0            | session   | engineer        | --              |
| Local_Netw...  | 1            | session   | --              | --              |

**4**

**5**

second  
↓

**New Policy**

☒ Add an existing policy **6** ☐ Create a new policy

Policy type: Session

Policy name: allowall **7**

Description:

Position:

**8** Cancel Submit

save  
→

**Pending Changes** **10**

**Roles** Policies Applications Aliases

**Roles 16**

| NAME                 | RULES    |
|----------------------|----------|
| denyall              | 1 Rules  |
| default-via-role     | 3 Rules  |
| default-vpn-role     | 4 Rules  |
| authenticated        | 4 Rules  |
| voice                | 45 Rules |
| Guest-99-guest-logon | 26 Rules |
| engineer             | 3 Rules  |

**9**

| engineer       | Policies     | Bandwidth | Captive Portal | Show Basic View |
|----------------|--------------|-----------|----------------|-----------------|
| NAME           | RULES COU... | TYPE      | POLICY US...   | DESCRIPTI...    |
| apprf-engin... | 0            | session   | engineer       | --              |
| engineer       | 0            | session   | engineer       | --              |
| Local_Netw...  | 1            | session   | engineer       | --              |
| allowall       | 2            | session   | default-iap... | --              |



# Role

## Assign role engineer to user local

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Auth Servers

AAA Profiles

L2 Authentication

Server Groups

| NAME          | SERVERS | FAIL THRO... | LOAD BAL... | SERVER RU... |
|---------------|---------|--------------|-------------|--------------|
| default       | 1       | --           | --          | 1            |
| internal      | 1       | --           | --          | 1            |
| 802.1X-G99... | 1       | --           | --          | 0            |

All Servers 1

| NAME     | TYPE | IP ADDRESS / ... | SERVER GROUP      |
|----------|------|------------------|-------------------|
| Internal | --   | --               | default intern... |

Server > Internal

Users

Options

Import

Guest user page

| USER NA... | ROLE        | EMAIL | EXPIRATI... | STATIC IP... | ENABLED |
|------------|-------------|-------|-------------|--------------|---------|
| a4d931h    | authenti... | --    | --          | 0.0.0.0      | Yes     |
| group99    | authenti... | --    | --          | 0.0.0.0      | Yes     |
| guest99    | guest       | --    | --          | 0.0.0.0      | Yes     |



Edit User group99

User name: group99

Password: \*\*\*\*\* 

Generate

Role: engineer

E-mail:

Static inner IP for RAPS: 0.0.0.0

Enabled: ☒

Expiration: None

Cancel

Submit

# Role

## Test connect WiFi > 802.1X-G99

ACCESS POINTS

1

0

CLIENTS

1

0

0

ALERTS

0

admin

Search

1

3

82.1 kB

2

Wireless Clients 1

| NAME    | IP ADDR...   | HEALTH   | BAND  | ROLE     |
|---------|--------------|----------|-------|----------|
| group99 | 192.168.10.2 | Unkno... | 5 GHz | engineer |

## Test ping gateway user

Command Prompt

Microsoft Windows [Version 10.0.19045.5965]  
(c) Microsoft Corporation. All rights reserved.

C:\Users\sakkr>ping 192.168.10.254

Pinging 192.168.10.254 with 32 bytes of data:  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.

Ping statistics for 192.168.10.254:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\sakkr>ping 192.168.10.253

Pinging 192.168.10.253 with 32 bytes of data:  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.

Ping statistics for 192.168.10.253:  
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

**CUSTOMER FIRST**  
**CUSTOMER LAST**

The image features the text "CUSTOMER FIRST" on the top line and "CUSTOMER LAST" on the bottom line. The letters are large, bold, and white. Each letter is filled with a different photograph of people at what appears to be a conference or event. The photos show various individuals, some smiling, some looking at devices, and others in group settings. The overall theme is customer-centricity.